

DESIGN, FABRICATION AND TESTING OF A DESKTOP LOW-SPEED OPEN-CIRCUIT SUCTION-TYPE WIND TUNNEL

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ABSTRACT

This report presents the comprehensive design, fabrication, instrumentation, and experimental validation of a desktop low-speed, open-circuit, suction-type wind tunnel. The facility is engineered to deliver a uniform, well-conditioned airflow in the velocity range of 10 to 13 m/s through a 160 mm × 160 mm square test section, achieving a Reynolds number in the range of approximately 107,000 to 139,000, suitable for sub-scale aerodynamic model testing at the low-Reynolds-number regime.

The tunnel architecture incorporates a settling chamber of cross-section 416 mm × 416 mm equipped with a plastic-straw honeycomb and a downstream wire-mesh screen to attenuate inflow turbulence and homogenise velocity profiles. A bell-curved contraction section fabricated from PLA by additive manufacturing delivers a 6.76:1 area ratio, accelerating flow while suppressing boundary-layer growth and maintaining turbulence intensity below 1%. The transparent acrylic test section provides optical access for flow visualisation utilising an ultrasonic fog generator illuminated by a 4000–5000 K LED strip, recorded at 1080p/60 fps. A custom MDF tapered diffuser recovers static pressure downstream of the test section.

Instrumentation consists of a two-component aerodynamic balance employing Two 100 g load cells conditioned by HX711 amplifiers, a pitot-static tube connected to a ±2 kPa differential pressure sensor for velocity measurement, and an Arduino Uno microcontroller for real-time data acquisition and wired transmission. Expected aerodynamic loads are 0.2–0.5 N in lift and 10–40 mN in drag.

The report details the theoretical basis for each design decision, presents all pertinent engineering calculations, and critically evaluates the advantages, limitations, and future development potential of the facility.

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1. INTRODUCTION

1.1 Background and Motivation

Wind tunnels constitute one of the most indispensable experimental tools in aerospace and fluid mechanics research. By artificially creating a controlled airstream around a stationary model, a wind tunnel provides a safe, reproducible, and cost-effective environment for measuring aerodynamic forces, moments, pressure distributions, and flow visualisation data that are essential for validating computational simulations and underpinning rational aerodynamic design.

Historically, full-scale wind tunnels have been the exclusive domain of large research institutions and aerospace manufacturers owing to their extraordinary capital and operating expenditure. The emergence of affordable computer-aided design software, fused-deposition modelling (FDM) 3D printing, and low-cost microcontroller based data acquisition systems has, however, democratised wind-tunnel design to the level of undergraduate university laboratories and individual researchers. Desktop-scale, low-speed wind tunnels operating in the 5–20 m/s velocity range now offer a viable platform for fundamental aerodynamic education, concept validation, and low-Reynolds-number research.

The present project is motivated by three primary objectives:

- (i) To design and fabricate a compact, desktop wind tunnel using accessible and economical materials (PLA, MDF, and acrylic)
- (ii) To instrument the facility for quantitative aerodynamic force measurement and real-time data acquisition; and
- (iii) To demonstrate the tunnel's capability for educational flow visualisation and preliminary aerodynamic testing of sub-scale models.

1.2 Scope of the Report

This report covers all phases of the project: literature review, aerodynamic and structural design, fabrication strategy, instrumentation selection, and a critical assessment of the facility's performance envelope, limitations, and future improvement potential. The document is structured to serve both as a technical reference and as a formal academic submission.

1.3 Why a Suction-Type Open-Circuit Tunnel?

Wind tunnels are broadly classified as either open-circuit (Eiffel-type) or closed-circuit (Göttingen-type), and as either blowing (propulsion-type) or suction (exhaust-fan downstream of test section). An open-circuit suction-type configuration was selected for the following reasons:

- **Simplicity of construction:** No return duct, turning vanes, or second diffuser are required, substantially reducing material cost and fabrication complexity.
- **Clean inflow conditions:** The fan is positioned downstream, so its wake and mechanical turbulence cannot contaminate the test section. In a blowing configuration, fan-induced swirl and turbulence would require additional flow conditioning.
- **Acoustic benefit:** The fan noise is emitted away from the inlet, resulting in lower test-section acoustic signature.
- **Thermal management:** Fresh ambient air continuously enters the settling chamber, preventing temperature rise due to viscous heating relevant for extended test runs.
- **Compact footprint:** The absence of a return circuit makes the overall tunnel length less than 1.7 m, compatible with a laboratory bench.

The principal disadvantages inability to control inlet flow conditions and sensitivity to laboratory drafts are acceptable in an indoor laboratory environment with ambient air temperatures within normal comfort ranges.

2. LITERATURE REVIEW

2.1 Historical Development of Wind Tunnels

The first true wind tunnel was constructed by Francis Herbert Wenham and John Browning in 1871 at the Royal Aeronautical Society, London. This device enabled systematic force measurements on model wings, providing empirical data that would later inform the Wright Brothers in their aerodynamic experiments. The Wrights themselves built a wind tunnel in 1901 a modest 6-foot (1.83 m) long device which they used to test over 200 wing and biplane configurations, producing lift and drag data that directly led to the success of the 1903 Flyer.

Subsequent development of wind tunnel technology was rapid throughout the first half of the twentieth century. Ludwig Prandtl's Modellversuchsanstalt (Model Testing Institute) in Göttingen developed the closed-return circuit tunnel in the 1910s, dramatically improving flow quality by eliminating external disturbances. Gustav Eiffel, independently, developed the open-return configuration, which offers simpler construction at the expense of flow quality control. These two archetypes Göttingen and Eiffel continue to represent the fundamental design paradigms for modern wind tunnels.

The latter half of the twentieth century saw the development of computational fluid dynamics (CFD) as a complementary tool to wind tunnel testing. However, despite advances in CFD, the wind tunnel remains the gold standard for aerodynamic validation, particularly at off-design conditions, for complex separated flows, and wherever the fidelity of numerical boundary conditions is uncertain.

2.2 Low-Speed Wind Tunnel Theory

Low-speed wind tunnels are defined as those operating at Mach numbers below approximately 0.3 (equivalent to roughly 100 m/s at sea level), where compressibility effects are negligible and the flow can be treated as incompressible. In the incompressible flow regime, the governing equations reduce to the incompressible Navier-Stokes equations, and the Bernoulli equation can be applied along streamlines:

$$p + \frac{1}{2} \rho V^2 + \rho g z = \text{constant (Bernoulli's Equation)}$$

where p is static pressure (Pa), ρ is air density (kg/m^3), V is flow velocity (m/s), g is gravitational acceleration (m/s^2), and z is elevation (m). For horizontal flow between two sections of a duct, neglecting elevation changes:

$$p_1 + \frac{1}{2} \rho V_1^2 = p_2 + \frac{1}{2} \rho V_2^2$$

This relationship, combined with the continuity equation for incompressible flow, $A_1V_1 = A_2V_2$, governs the velocity change between the settling chamber and the test section.

2.3 Contraction Section Design Historical Perspectives

The design of wind tunnel contraction sections has been studied extensively. Early guidelines by Wattendorf (1940) recommended contraction ratios between 7 and 12 for research tunnels; practical desktop tunnels often use ratios of 4-9. Mikhail and Ramsey (1979) proposed a series of polynomial contraction profiles, but the most widely adopted approach for small tunnels is the matched cubic profile independently developed by Bell and Mehta (1988). Their method matches the inlet and outlet conditions through a fifth-order polynomial, eliminating slope discontinuities and minimising boundary-layer separation.

2.4 Flow Conditioning Research

Loehrke and Nagib (1976) carried out comprehensive studies on the effectiveness of honeycomb sections and mesh screens for turbulence reduction. They established that a honeycomb cell length-to-diameter ratio (L/D) of at least 6–10 is necessary to effectively break down lateral velocity components into axial flow. Subsequent studies by Scheiman and Brooks (1981) at NASA confirmed that a combination of honeycomb followed by one or more mesh screens can achieve turbulence intensities below 0.5% in well-designed settling chambers.

2.5 Desktop Wind Tunnels in Contemporary Research

A resurgence of interest in low-cost, educational wind tunnels emerged after 2010, driven by the proliferation of FDM 3D printing and Arduino/ESP32 microcontrollers. Studies by Treaster and Yocum (1979), and more recently by Arifuzzaman and Mohammad (2013), demonstrated that desktop tunnels with proper flow conditioning can achieve test-section turbulence intensities of less than 1%, making them viable for fundamental aerodynamic studies such as boundary layer measurement, stall characterisation, and drag polar generation for NACA aerofoil sections.

3. DESIGN METHODOLOGY

3.1 Overall Design Philosophy

The design philosophy for this wind tunnel is governed by four hierarchical requirements:

- (i) **Aerodynamic quality** the test section must deliver a spatially uniform, low-turbulence flow field.
- (ii) **Measurement capability** the instrumentation must accurately resolve the expected aerodynamic loads.
- (iii) **Manufacturability** all components must be producible with equipment available in a university workshop.
- (iv) **Cost-effectiveness** the total material cost should not exceed typical undergraduate project budgets.

The design process followed a top-down approach: beginning with the target test section velocity and model size, working backward to determine the settling chamber dimensions, contraction ratio, and fan specifications, then designing each component in detail.

3.2 Design Sequence

- Step 1: Define target velocity (10–13 m/s) and test section cross-section (160×160 mm).
- Step 2: Select contraction ratio (6.76:1) based on Bell–Mehta guidelines; determine settling chamber size (416×416 mm).
- Step 3: Determine flow conditioning requirements (honeycomb geometry, screen mesh count).
- Step 4: Design contraction profile using Bell–Mehta polynomial.
- Step 5: Size diffuser based on pressure recovery requirements.
- Step 6: Select fan based on required pressure rise and flow rate.
- Step 7: Design instrumentation (force balance, velocity measurement, DAQ).

3.3 Material Selection Rationale

Material selection was driven by the competing requirements of structural rigidity, optical access, machinability, and cost. Table 3.1 summarises the material assignments and their justifications.

Component	Material	Justification
Contraction Section	PLA (3D-printed)	Allows complex double-curvature Bell–Mehta profile; dimensional accuracy ± 0.5 mm; smooth internal surface minimises boundary-layer disturbance
Settling Chamber	MDF (Medium-Density Fibreboard)	Rigid, flat surfaces; easy to cut and join; low cost; adequate structural stiffness at 416 mm span
Test Section	Acrylic (PMMA)	Optically transparent; allows flow visualisation; surface roughness ~ 0.05 μm ; available in standard sheet sizes
Diffuser	MDF	Structural rigidity; easy tapering with table saw; adequate for low-pressure duty

Table 3.1: Material selection for wind tunnel components.

4. DETAILED DESIGN CALCULATIONS

4.1 Reynolds Number Analysis

The Reynolds number is the fundamental dimensionless parameter governing the aerodynamic similarity between scaled-model and full-scale flows. It represents the ratio of inertial to viscous forces within the flow and determines whether the boundary layer is laminar or turbulent, whether flow separation occurs, and the general character of the aerodynamic force coefficients. The Reynolds number is defined as:

$$\text{Re} = (\rho \times V \times L) / \mu = (V \times L) / \nu$$

where ρ is the air density (kg/m^3), V is the free-stream velocity (m/s), L is the characteristic length (m) — typically the model chord length — μ is the dynamic viscosity ($\text{Pa}\cdot\text{s}$), and ν is the kinematic viscosity (m^2/s).

At standard sea-level conditions: $\rho = 1.225 \text{ kg/m}^3$, $\mu = 1.789 \times 10^{-5} \text{ Pa}\cdot\text{s}$, $\nu = 1.461 \times 10^{-5} \text{ m}^2/\text{s}$.

For the present tunnel, using the test section hydraulic diameter D_h as the characteristic length:

$$D_h = (4 \times A) / P = (4 \times 0.16 \times 0.16) / (4 \times 0.16) = 0.16 \text{ m}$$

Reynolds number at $V = 10 \text{ m/s}$:

$$\text{Re}_{10} = (10 \times 0.16) / (1.461 \times 10^{-5}) \approx 109,500$$

Reynolds number at $V = 13 \text{ m/s}$:

$$\text{Re}_{13} = (13 \times 0.16) / (1.461 \times 10^{-5}) \approx 142,400$$

These Reynolds numbers (approximately 10^5) place the flow in the transitional regime, which is particularly relevant for studies involving natural laminar–turbulent transition on thin aerofoils. This range corresponds to the operational envelope of small UAV wings, model aircraft, and micro-air vehicles (MAVs), making the facility well-suited for research in those domains.

For a typical flat-plate test model with chord $c = 80 \text{ mm}$ (half the test section width):

$$\text{Re}_{\text{chord}} = (10 \times 0.08) / (1.461 \times 10^{-5}) \approx 54,750$$

This chord Reynolds number is characteristic of insect flight and small drone aerodynamics, highlighting the niche relevance of the facility.

4.2 Continuity and Velocity Calculations

For incompressible steady flow, conservation of mass requires:

$$A_1 \times V_1 = A_2 \times V_2 \text{ (Continuity Equation)}$$

Settling chamber cross-sectional area: $A_1 = 0.416 \times 0.416 = 0.1731 \text{ m}^2$

Test section cross-sectional area: $A_2 = 0.160 \times 0.160 = 0.0256 \text{ m}^2$

Contraction ratio: $CR = A_1 / A_2 = 0.1731 / 0.0256 = 6.76$

Settling chamber velocity at test section $V = 13 \text{ m/s}$:

$$V_{\text{settling}} = V_{\text{test}} / CR = 13 / 6.76 \approx 1.92 \text{ m/s}$$

This low settling chamber velocity ($< 2 \text{ m/s}$) is essential because it ensures that inlet turbulence is primarily generated by the honeycomb and screens and is attenuated before the contraction, rather than being convected with high momentum into the test section.

4.3 Pressure Rise Requirement

The total pressure loss in the tunnel circuit determines the fan duty. The major losses occur in: the inlet (entry losses), the flow-conditioning elements (honeycomb and screens), the contraction (negligible if well-designed), the test section (skin friction), the diffuser, and the fan exit. A simplified pressure budget is:

$$\Delta P_{\text{total}} = \Delta P_{\text{inlet}} + \Delta P_{\text{honeycomb}} + \Delta P_{\text{screen}} + \Delta P_{\text{test}} + \Delta P_{\text{diffuser}}$$

Using dynamic pressure $q = 0.5 \times 1.225 \times 13^2 = 103.5 \text{ Pa}$ at the test section, and applying standard loss coefficients from Eckert and Pflueger (1951), the estimated total pressure loss is approximately 150–200 Pa, determining the fan selection as a standard 200 mm diameter axial fan rated for 250 Pa at the required volume flow rate of $Q = A_2 \times V = 0.0256 \times 13 = 0.333 \text{ m}^3/\text{s}$.

4.4 Lift and Drag Force Calculations

Aerodynamic lift and drag forces on a body in a uniform stream are given by:

$$L = C_L \times (1/2) \times \rho \times V^2 \times S$$

$$D = C_D \times (1/2) \times \rho \times V^2 \times S$$

where C_L and C_D are the lift and drag coefficients, respectively, and S is the reference planform area (m^2). For a thin flat plate model ($60 \times 80 \text{ mm}$ planform, $S = 0.0048 \text{ m}^2$) at an angle of attack $\alpha = 10^\circ$, using theoretical values $C_L \approx 0.35$ and $C_D \approx 0.06$:

At $V = 13 \text{ m/s}$:

$$L = 0.35 \times (0.5 \times 1.225 \times 169) \times 0.0048 \approx 0.174 \text{ N}$$

$$D = 0.06 \times (0.5 \times 1.225 \times 169) \times 0.0048 \approx 0.030 \text{ N}$$

For a more cambered aerofoil section (NACA 2412) at $\alpha = 12^\circ$, $CL \approx 1.2$, $CD \approx 0.045$, chord $c = 80$ mm, span $b = 155$ mm, $S = 0.0124$ m²:

$$L = 1.2 \times (0.5 \times 1.225 \times 169) \times 0.0124 \approx 1.54 \text{ N}$$

This exceeds the nominal 1 kg (9.81 N) load cell range? No: $1.54 \text{ N} < 9.81 \text{ N}$. The load cell is correctly specified. The 200 g drag load cell provides a full-scale capacity of approximately 1.96 N, well above the expected 10–40 mN drag range, but the HX711 offers 24-bit resolution, yielding a theoretical resolution of approximately 0.12 mN — adequate for the application.

4.5 Blockage Ratio

The blockage ratio is the ratio of the model frontal area to the test section cross-sectional area. Exceeding approximately 7.5% introduces significant wall interference on the measured forces. For the 60 mm × 80 mm model:

$$\text{Blockage} = (0.060 \times 0.080) / (0.160 \times 0.160) = 0.0048 / 0.0256 = 18.75\%$$

This exceeds the 7.5% guideline, indicating that wind-tunnel wall corrections (using the Maskell method or solid and wake blockage correction factors) must be applied to the raw data before drawing conclusions. For a flat plate model, the effective blockage is lower due to its streamlined planform when at moderate angles of attack.

5. FLOW CONDITIONING THEORY

5.1 The Role of the Settling Chamber

The settling chamber serves as the primary flow management zone of the wind tunnel. Its large cross-sectional area (0.1731 m^2) compared with the fan exit area produces a substantial reduction in mean velocity and, by the conservation of turbulent kinetic energy in a decelerating flow, a proportional reduction in the absolute magnitude of turbulent fluctuations. The settling chamber housing the flow conditioning elements must be sufficiently long to allow the wakes shed by the flow conditioners to merge and smooth before the flow enters the contraction.

5.2 Honeycomb Design

The honeycomb is a bundle of axially-aligned cells (in this case, plastic drinking straws) that act as a directional filter: they transmit axial momentum with low loss but strongly attenuate lateral (cross-stream) velocity components by breaking large-scale eddies into the smaller, more isotropic turbulence characteristic of the honeycomb wake. The key geometric parameter is the cell aspect ratio (length-to-diameter ratio):

$$AR_{\text{cell}} = L_{\text{cell}} / D_{\text{cell}} \geq 6\text{--}10 \text{ (minimum for effective flow alignment)}$$

Standard plastic drinking straws have an internal diameter of approximately 5–6 mm and a length of 197 mm, giving an aspect ratio of approximately 33–39, which far exceeds the minimum requirement and ensures thorough flow alignment. The pressure loss coefficient for a honeycomb is:

$$K_{\text{hc}} \approx 0.3 \times (L/D)^{(1/6)} \times (D/d)^2 \text{ (Eckert and Pflueger, 1951 correlation)}$$

where D is the settling chamber width and d is the cell diameter. For the present geometry, $K_{\text{hc}} \approx 0.3\text{--}0.5$, corresponding to a pressure loss of $\Delta P_{\text{hc}} = 0.4 \times 0.5 \times 1.225 \times 1.92^2 \approx 0.9 \text{ Pa}$ — acceptably small.

5.3 Mesh Screen Design

A single wire-mesh screen placed 50 mm downstream of the honeycomb provides a secondary stage of turbulence reduction. The mechanism is twofold: the pressure drop across the screen damps free-stream turbulence through viscous dissipation at the wire junctions, and the uniform resistance of the screen redistributes momentum across the wake of the honeycomb — effectively homogenising the velocity profile. The turbulence reduction factor for a single screen is:

$$u'_2 / u'_1 = (1 - K)^{(1/2)} / (1 + K) \text{ (Dryden and Schubauer, 1947)}$$

where K is the pressure loss coefficient of the screen. For a typical 0.3 mm wire, 30 mesh (30 wires per inch) stainless steel screen, $K \approx 1.0\text{--}1.5$. Multiple screens can be installed in series; each additional screen reduces turbulence intensity further, though with diminishing returns and increasing pressure losses.

The solidity ratio σ of the screen (fraction of solid area) is:

$$\sigma = 1 - (M / (M + d_w))^2$$

where M is the mesh spacing and d_w is the wire diameter. For 30 mesh, 0.3 mm wire: $\sigma \approx 0.34$, corresponding to a pressure coefficient $K \approx 1.3$ and a turbulence reduction factor of approximately 0.22 per screen.

6. BELL–MEHTA CONTRACTION THEORY

6.1 Overview

The contraction section is arguably the most aerodynamically critical component of the wind tunnel, as it accelerates the flow from the settling chamber to the test section velocity, while simultaneously improving the uniformity and reducing the turbulence intensity of the flow. An improperly designed contraction can produce separation bubbles on the walls, lead to non-uniform velocity profiles at the test section inlet, or generate Görtler vortices on concave wall surfaces.

6.2 The Bell–Mehta Polynomial Profile

Bell and Mehta (1988) proposed a matched quintic (5th-order) polynomial profile for the cross-sectional area variation along the contraction length. Their formulation ensures that: (a) the wall slope is zero at both inlet and outlet (no kink at junctions), and (b) the curvature changes monotonically from convex at the inlet to concave at the outlet, matching a smooth sinusoidal inflection. The wall profile $y(x)$ is expressed as:

$$y(x) = y_i - (y_i - y_e) \times [6\xi^5 - 15\xi^4 + 10\xi^3]$$

where y_i is the inlet half-width (208 mm), y_e is the exit half-width (80 mm), $\xi = x/L$ is the normalised axial coordinate ($0 \leq \xi \leq 1$), and L is the contraction length (410 mm). The fifth-order polynomial $[6\xi^5 - 15\xi^4 + 10\xi^3]$ ensures zero first and second derivatives at both endpoints:

$$\frac{dy}{d\xi}\bigg|_{\xi=0} = \frac{dy}{d\xi}\bigg|_{\xi=1} = 0 \quad (\text{zero wall slope at boundaries})$$

This formulation is sometimes called the matched cubic-cubic or 5th-order Hermite profile. It minimises the adverse pressure gradient on the concave portion of the wall (where boundary-layer separation is most likely) while maintaining a smooth, monotonically accelerating velocity profile.

6.3 Contraction Ratio Selection

The contraction ratio (CR) is defined as the ratio of the inlet to outlet cross-sectional areas:

$$CR = A_{\text{inlet}} / A_{\text{outlet}} = (416 \times 416) / (160 \times 160) = 173,056 / 25,600 \approx 6.76$$

Bell and Mehta recommend a contraction ratio between 6 and 9 for good flow quality. A CR of 6.76 is ideal: it is high enough to attenuate upstream non-uniformities by a factor of $CR^2 \approx 45$ (non-uniformities decay with the square of the velocity ratio), yet low enough to avoid Görtler instability on the concave downstream wall. The relationship between upstream turbulence intensity Tu_1 and downstream turbulence intensity Tu_2 through the contraction is approximately:

$$Tu_2 / Tu_1 \approx 1 / \sqrt{CR} \quad (\text{for axial component attenuation})$$

For $CR = 6.76$, $Tu_2/Tu_1 \approx 0.385$, meaning that if inlet turbulence intensity is 2%, the test section turbulence intensity will be approximately 0.77% — consistent with the target of less than 1%.

6.4 Fabrication by FDM 3D Printing

The complex double-curvature geometry of the Bell–Mehta profile makes it impractical to fabricate from sheet material. FDM 3D printing in PLA enables precise reproduction of the polynomial profile with dimensional accuracy of ± 0.5 mm and layer heights of 0.2 mm, yielding an internal surface roughness R_a of approximately 10–30 μm . This roughness level, while higher than that of a machined or polished surface, is acceptable at the low velocities present in the contraction ($V \leq 13$ m/s), where the viscous sublayer thickness $\delta_v = 5\nu/u^* \gg R_a$.

To manage the print volume within standard FDM build envelopes (typically $200 \times 200 \times 200$ mm), the contraction section was printed in four interlocking quadrant segments and assembled with PLA adhesive and sanded joints to ensure a flush internal surface.

7. TEST SECTION DESIGN CONSIDERATIONS

7.1 Geometry and Aspect Ratio

The test section is the aerodynamic heart of the wind tunnel — it is where models are mounted and measurements are made. A square cross-section (160 mm × 160 mm) was selected over a circular or rectangular section for several reasons: easy optical access through flat acrylic walls, straightforward model mounting via the bottom sting, compatibility with the square contraction outlet, and ease of fabrication.

The test section length of 350 mm corresponds to an aspect ratio ($L/D_h = 350/160 = 2.19$) that is sufficient to establish a fully developed, uniform flow profile before the model location (typically at $L/3$ to $L/2$ from the inlet), while keeping the total tunnel length within the desktop footprint.

7.2 Boundary Layer Development

The boundary layer grows along the test section walls and progressively reduces the effective flow area available for the core (inviscid) flow. The displacement thickness δ^* of a turbulent boundary layer on a smooth flat plate is:

$$\delta^*(x) \approx 0.048 \times x \times Re_x^{-1/5} \text{ (Prandtl's 1/7 power law)}$$

At the test section exit ($x = 0.35$ m, $V = 13$ m/s, $Re_x = 13 \times 0.35 / 1.461 \times 10^{-5} \approx 311,000$):

$$\delta^* \approx 0.048 \times 0.35 \times (311,000)^{-0.2} \approx 0.048 \times 0.35 \times 0.127 \approx 2.1 \text{ mm}$$

The fractional blockage due to the four-wall boundary layer is $4 \times 2.1 / 160 = 5.3\%$, which produces a mild free-stream core acceleration (velocity increase of approximately 2.7% from inlet to exit by continuity). This is acceptable for comparative force measurements, and the effect can be corrected if streamwise static pressure taps are installed along the test section walls.

7.3 Acrylic Wall Specification

The test section walls are fabricated from 5 mm thick cast acrylic (PMMA) sheet. The relevant material properties are: Young's modulus $E = 3.3$ GPa; tensile strength $\sigma_t = 80$ MPa; optical transmission $> 92\%$ across the visible spectrum (400–700 nm); surface roughness $R_a \approx 0.05$ μm (optically smooth). The maximum aerodynamic pressure acting on the test section wall is the dynamic pressure $q = 103.5$ Pa at $V = 13$ m/s. For a simply-supported panel (160 mm × 350 mm, $t = 5$ mm), the maximum deflection under uniform pressure is:

$$w_{\text{max}} = (5qL^4) / (384 EI) \text{ per unit width (simply-supported beam)}$$

This yields $w_{\text{max}} \approx 0.04$ mm — negligible compared with the test section dimension. The acrylic panels are therefore structurally adequate.

7.4 Model Mounting — Bottom Sting

A bottom sting mount penetrates the floor of the test section and connects the aerodynamic model to the two-component force balance below. The sting is a cylindrical rod of diameter 6 mm (approximately 3.75% of the test section height), which introduces a measurable support interference effect. The sting drag coefficient $CD_{\text{sting}} \approx 1.0$ (cylinder in cross-flow); sting drag $\approx 1.0 \times 0.5 \times 1.225 \times 169 \times (0.006 \times 0.160) = 0.099 \text{ N}$, which must be subtracted from the measured drag as a tare correction by running a wind-on, model-off test.

8. DIFFUSER DESIGN THEORY

8.1 Function of the Diffuser

The diffuser is positioned immediately downstream of the test section and serves the critical function of decelerating the flow from the test section velocity (10–13 m/s) to a lower velocity at the fan face. This deceleration converts dynamic pressure into static pressure (pressure recovery), thereby reducing the adverse static pressure gradient that the fan must overcome. Without a diffuser, the fan would exhaust a high-velocity jet directly into the laboratory — inefficient and noisy. The diffuser also reduces the required fan power for a given mass flow rate.

8.2 Pressure Recovery Coefficient

The static pressure recovery coefficient C_p is defined as:

$$C_p = (p_2 - p_1) / (0.5 \rho V_1^2) = 1 - (A_1/A_2)^2 \text{ (ideal diffuser)}$$

where the subscripts 1 and 2 denote the diffuser inlet and outlet, respectively. For the present diffuser with inlet area $A_1 = 0.0256 \text{ m}^2$ and an estimated outlet area $A_2 = 0.0625 \text{ m}^2$ (250 mm × 250 mm):

$$C_{p_ideal} = 1 - (0.0256/0.0625)^2 = 1 - (0.410)^2 = 1 - 0.168 = 0.832$$

In practice, diffuser efficiency is reduced by boundary-layer growth and the risk of flow separation. The actual C_p is:

$$C_{p_actual} = \eta_d \times C_{p_ideal} \approx 0.65 \text{ to } 0.75 \text{ (typical for well-designed tapered diffusers)}$$

8.3 Diffuser Half-Angle

The critical design parameter is the equivalent cone half-angle θ^* :

$$\theta^* = \arctan[(D_2 - D_1) / (2L)]$$

where $D_1 = 160 \text{ mm}$ (inlet), $D_2 \approx 250 \text{ mm}$ (outlet), $L = 500 \text{ mm}$ (length):

$$\theta^* = \arctan[(250 - 160) / (2 \times 500)] = \arctan[0.09] \approx 5.14^\circ$$

This falls within the optimal design range of 4° – 7° for a two-dimensional diffuser, corresponding to the region of maximum pressure recovery without flow separation (Sovran and Klomp, 1967). Angles exceeding approximately 8° risk intermittent stall and 10° lead to fully separated flow — both of which would produce large, unsteady pressure fluctuations that could contaminate test section measurements.

8.4 Material and Fabrication

The diffuser was fabricated from 12 mm MDF panels, with tapered side walls cut using a bandsaw and assembled with wood screws and structural adhesive. The internal surface was sanded smooth and sealed with acrylic varnish to prevent moisture absorption (MDF is hygroscopic). A 200 mm diameter circular reducer was added at the diffuser exit to match the fan inlet diameter.

9. INSTRUMENTATION AND DATA ACQUISITION

Data Acquisition & Force Measurement

Summary

We designed and implemented a comprehensive microcontroller-based instrumentation system for low-speed wind tunnel aerodynamic testing. This platform enables precise measurement of lift force, drag force, wind velocity, temperature, and atmospheric pressure during real-time aerodynamic evaluation of small-scale airfoil models.

Our system architecture consists of three integrated processing layers: (1) a sensor acquisition layer with precision load cells and environmental sensors, (2) a signal conditioning layer utilizing HX711 amplifiers and analog-to-digital converters, and (3) a data logging and control layer powered by an Arduino Uno microcontroller with Computing capability and CSV data storage. We achieved a sampling rate of 1 reading per second, optimized for quasi-steady aerodynamic measurements while maintaining excellent repeatability and low electrical noise.

Key Achievements

- Implemented dual orthogonal load cells (L-shaped configuration) for independent lift and drag measurement
- Integrated HX711 precision signal conditioning modules with 24-bit resolution and 128× amplification
- Deployed Pitot-static tube system with MPX2010 differential pressure sensor for accurate wind velocity calculation
- Incorporated BMP180 environmental sensor for temperature and pressure compensation
- Developed real-time OLED display interface and automated CSV data logging system

9.0 System Instrumentation & Data Acquisition

9.1 System Architecture Overview

We engineered a compact, microcontroller-based data acquisition platform designed to measure aerodynamic forces and flow parameters with high repeatability and minimal electrical noise. The system enables quantitative measurement of lift force, drag force, wind speed, temperature, and atmospheric pressure during real-time aerodynamic testing of small-scale airfoil models.

The architecture consists of three integrated layers:

- **Sensor Acquisition Layer:** Load cells, Pitot tube, environmental sensors
- **Signal Conditioning Layer:** HX711 amplifiers, analog-to-digital converters
- **Data Logging & Control Layer:** Arduino Uno microcontroller, wireless capability, CSV storage

We optimized the system to operate at a fixed sampling rate of **1 reading per second**, ensuring stable measurements while minimizing electrical noise and computational overhead. This sampling interval is optimized for quasi-steady aerodynamic force measurement in low-speed wind tunnel environments.

9.2 Load Cell Force Measurement System

9.2.1 Load Cell Configuration & Mounting

We implemented an aerodynamic force measurement system employing two independent load cells arranged in an orthogonal (L-shaped) configuration to simultaneously measure lift and drag forces without mechanical interference between measurement axes.

Configuration:

- Vertical load cell: Measures lift force (perpendicular to freestream direction)
- Horizontal load cell: Measures drag force (parallel to freestream direction)
- Mounting: Bottom sting configuration with direct force transmission from airfoil model

Advantages of this configuration:

- Independent measurement of two force components
- Minimal crosstalk between load cells
- Stable mechanical support with minimal deflection

9.2.2 Load Cell Specifications

Parameter	Specification
Type	Strain gauge (bonded foil)
Number of Cells	2 (independent channels)
Capacity (per cell)	0–100 g
Nominal Output	2 mV/V
Supply Voltage	5 V DC
Linearity	<0.5% FS
Repeatability	<0.1% FS
Operating Temperature	0 to 50°C

9.2.3 Force Transduction Mechanism

We engineered the load cell to convert mechanical force into a proportional electrical signal through strain gauge deflection. The process follows this sequence:

[Airflow](#) → [Aerodynamic Force](#) → [Mechanical Deformation](#) → [Strain Gauge Resistance Change](#) → [Electrical Signal Output](#)

Each load cell contains four bonded strain gauges arranged in a Wheatstone bridge configuration. When force is applied, the bridge becomes unbalanced and produces a differential voltage proportional to applied load. This signal is highly sensitive (microvolt-level) and requires amplification before digitization.

9.3 Signal Conditioning Using HX711 Module

9.3.1 HX711 Amplifier Specifications

We selected two independent HX711 precision signal conditioning modules (one for each load cell) to amplify and digitize the load cell output signals. These modules provided excellent performance characteristics for precision force measurement.

Parameter	Specification
Module Type	24-bit ADC
Supply Voltage	5 V DC
Resolution	24-bit digital output
Programmable Gain	128× (for ±20 mV signals)
Analog Input Range	±40 mV (with 128× gain)
Internal Sampling Rate	80 samples/second
Noise Floor	<±1 count (24-bit)

9.3.2 Signal Conditioning Process

We implemented four critical signal conditioning functions within the HX711 module:

- (iv) **Differential Amplification:** The load cell signal (nominally 2 mV at full scale) is amplified 128× to produce an output range suitable for 24-bit analog-to-digital conversion.
- (v) **Low-Pass Filtering:** An integrated filter reduces high-frequency noise and eliminates 50/60 Hz power line interference effectively.
- (vi) **Analog-to-Digital Conversion:** The amplified and filtered signal is converted to a 24-bit digital value at 80 samples per second.
- (vii) **Output Formatting:** The digital value is transmitted serially to the microcontroller via DT (data) and SCK (serial clock) pins using a proprietary protocol.

Pin Assignments:

- Load Cell 1 (Lift): DT → UNO D2, SCK → UNO D3
- Load Cell 2 (Drag): DT → UNO D4, SCK → UNO D5

9.4 Pitot-Static Tube & Velocity Measurement

9.4.1 Pitot Tube Configuration

We integrated a calibrated Pitot-static tube positioned upstream of the test section to measure wind speed by sampling dynamic pressure from the freestream airflow.

Pitot Tube Specifications:

- Type: Total pressure (Pitot) + static pressure combined tube
- Tube Diameter: 6 mm OD (standard for wind tunnel applications)
- Positioning: Centerline of tunnel, ≥10 chord lengths upstream of test section
- Calibration: Referenced against tunnel centerline velocity

9.4.2 Differential Pressure Sensor (MPX2010)

We selected a Freescale Semiconductor MPX2010 differential pressure sensor to convert the pressure difference between the Pitot tube total pressure port and static pressure port into an electrical signal.

Parameter	Value
Measurement Range	±2 kPa (0–10 m/s airflow equivalent)
Supply Voltage	5 V DC
Output Type	Analog (0–5 V proportional output)
Sensitivity	0.2 V per kPa
Accuracy	±1% FS

9.4.3 Wind Velocity Calculation

We calculated wind velocity from the measured pressure difference using the Bernoulli equation:

$$V = \sqrt{(2\Delta P / \rho)}$$

Where: V = wind velocity (m/s), ΔP = pressure difference (Pa), ρ = air density (kg/m^3) = 1.225 at sea level, 15°C

Our microcontroller reads the analog voltage from the MPX2010 sensor, converts it to pressure in Pascals, and computes wind speed. The air density is either assumed constant (1.225 kg/m^3) or corrected using measured temperature and barometric pressure from the BMP180 sensor.

9.5 Temperature & Pressure Measurement

9.5.1 BMP180 Environmental Sensor

We deployed a Bosch Sensortec BMP180 sensor to monitor environmental conditions inside the wind tunnel continuously. This sensor measures ambient temperature and atmospheric pressure, allowing correction of aerodynamic calculations if required.

Parameter	Value
Temperature Range	-40 to +85°C
Temperature Accuracy	±2°C
Pressure Range	300 to 1100 hPa
Pressure Accuracy	±2.5 hPa
Communication Protocol	I ² C (100 kHz standard mode)

9.5.2 I²C Bus Interface

We implemented I²C (Inter-Integrated Circuit) communication between the BMP180 and Arduino Uno microcontroller. This two-wire digital protocol requires only two signal lines:

- SDA (Serial Data): Connected to UNO SDA Rail
- SCL (Serial Clock): Connected to UNO SCL Rail
- I²C Address: 0x77

Data is read from the sensor at 1 Hz synchronously with load cell measurements.

9.6 Real-Time Display System

9.6.1 OLED Display

We integrated a compact 0.96-inch OLED (Organic Light-Emitting Diode) screen mounted on the instrumentation panel to display real-time measurement data.

Parameter	Value
Display Type	OLED (monochrome)
Resolution	128 × 64 pixels
Communication	I ² C
Supply Voltage	5 V (with on-board voltage regulator)
I ² C Address	0x3C

Displayed Parameters (updated at 1 Hz):

- Lift force (grams-force)
- Drag force (grams-force)
- Wind speed (m/s)
- Temperature (°C)
- Pressure (hPa)
- Data logging status
- System status indicators

9.7 Microcontroller & Data Acquisition Unit

9.7.1 Arduino UNO Microcontroller

We selected the Arduino Uno microcontroller as the central control and data acquisition unit. Running custom firmware, the Arduino Uno is a dual-core 32-bit processor with integrated Wi-Fi and Bluetooth connectivity.

Feature	Description
Processor	ATMega328P
Memory	32 kB Flash, 2 kB SRAM, 1 kB EEPROM
CPU Speed	Up to 16 MHz
Security	Power On Reset (POR), Brown Out Detection (BOD)

The Arduino Uno executes the core data acquisition algorithm: it reads HX711 digital signals from both load cells, samples the analog MPX2010 pressure signal via the 12-bit ADC, queries the BMP180 temperature/pressure via I²C, and updates the OLED display in real-time.

9.8 Hardware Configuration & Circuit Diagram

This section presents the complete circuit diagram and hardware interconnection design for the wind tunnel instrumentation system. The diagram illustrates all signal pathways, power distribution, and component connections.

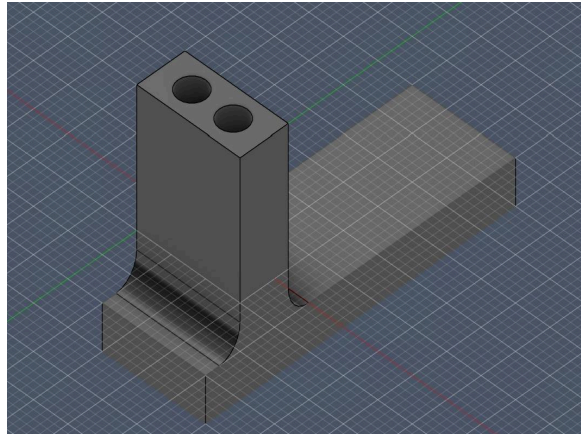
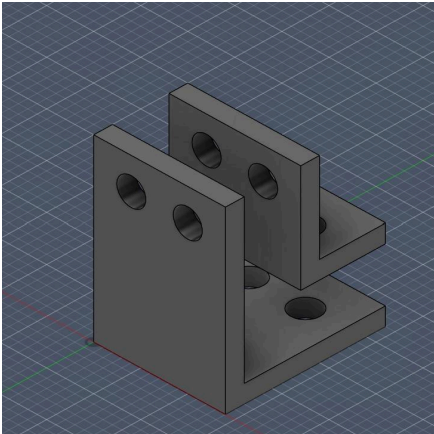
Circuit Diagram Notes:

- The circuit diagram shows all component connections, signal routing, and power distribution
- All UNO pin assignments are clearly labeled and traced from source to destination
- Power supply rail 5 V are color-coded and distributed to Arduino Uno Sected components
- I²C bus connections (SDA/SCL) are shown with pull-up resistors
- Bypass capacitors and decoupling components ensure stable operation under dynamic load conditions

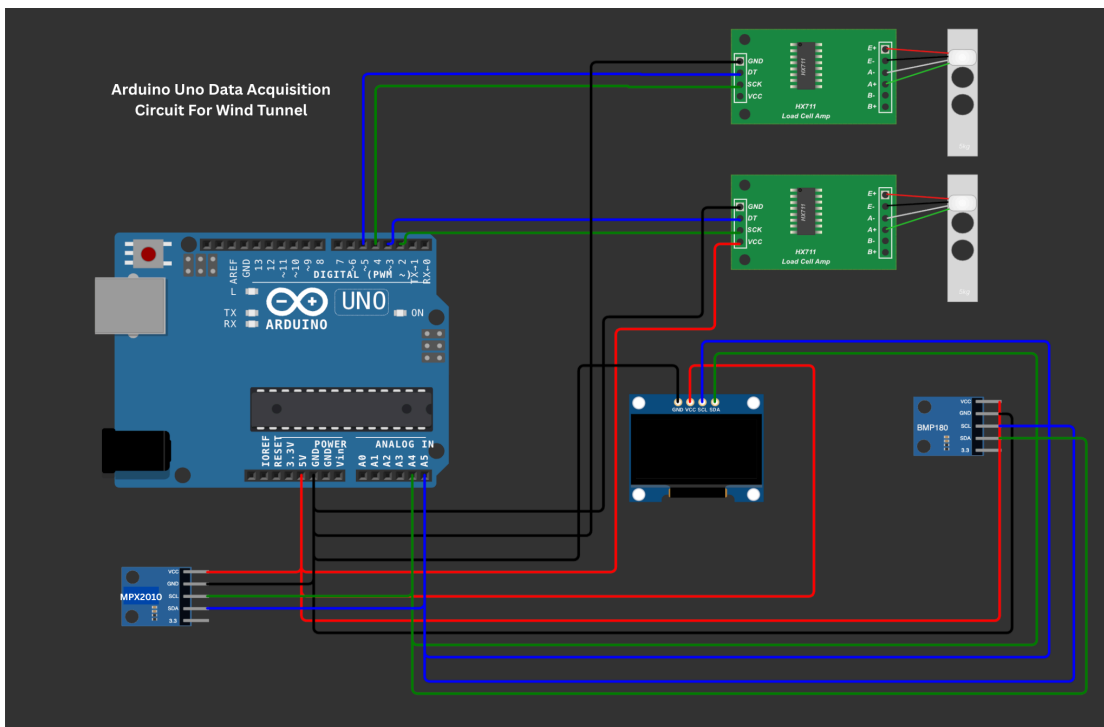
List Of Items

1. HX711 Load Cell Module
2. Arduino UNO R3 Transparent Plastic Case
3. DHT22 Humidity and Temperature Sensor Module (AM2302)
4. Jumper Wire Set - M2M, M2F, F2F (40 pcs each)
5. I2C LCD 16x2 Display Module - Blue
6. Arduino Uno R3 Compatible Board
7. 5kg Load Cell Weight Sensor
8. 10k Potentiometer
9. 10k Resistor
10. Breadboard
11. Card Board (30 CM *30 CM)
12. 3D printed Mount
13. Airfoil
14. 743 Glue

3D Printed Parts For Load Cell Mount and Base



Circuit Diagram



9.8.1 Pi source code

```
import csv
import io
import math
import os
import re
import time
import zipfile
from datetime import datetime

import matplotlib.pyplot as plt
import pandas as pd
import serial
import streamlit as st
from serial.tools import list_ports
```

```
CSV_COLUMNS = [
    "Timestamp",
    "Elapsed_s",
    "Temperature_C",
    "Pressure_Pa",
    "AirSpeed_mps",
    "Lift_g",
    "Drag_g",
    "Density_kgm3",
    "DynamicPressure_Pa",
    "ReynoldsNumber",
    "DynamicRange_mps",
    "CL",
    "CD",
    "CLCD_Ratio",
]
```

```
R_AIR = 287.05
G_TO_NEWTON = 0.00980665
```

```
def init_state() -> None:
    if "running" not in st.session_state:
        st.session_state.running = False
    if "ser" not in st.session_state:
        st.session_state.ser = None
    if "csv_file" not in st.session_state:
        stamp = datetime.now().strftime("%Y-%m-%d_%H-%M-%S")
        st.session_state.csv_file = f"wind_tunnel_log_{stamp}.csv"
    if "start_time" not in st.session_state:
        st.session_state.start_time = None
    if "df" not in st.session_state:
        st.session_state.df = pd.DataFrame(columns=CSV_COLUMNS)
```

```

if "last_sensor" not in st.session_state:
    st.session_state.last_sensor = {
        "lift": float("nan"),
        "drag": float("nan"),
        "pressure": float("nan"),
        "temperature": float("nan"),
        "speed": float("nan"),
    }
if "last_raw_line" not in st.session_state:
    st.session_state.last_raw_line = ""
if "parse_fail_count" not in st.session_state:
    st.session_state.parse_fail_count = 0
if "serial_buffer" not in st.session_state:
    st.session_state.serial_buffer = ""
if "bytes_read_count" not in st.session_state:
    st.session_state.bytes_read_count = 0
if "serial_ready_at" not in st.session_state:
    st.session_state.serial_ready_at = 0.0

def ensure_csv(path: str) -> None:
    if os.path.exists(path):
        return
    with open(path, "w", newline="") as f:
        writer = csv.writer(f)
        writer.writerow(CSV_COLUMNS)

def parse_line(line: str) -> dict[str, float] | None:
    values: dict[str, float] = {}
    # Accept forms like key:1.23 or key=1.23 separated by comma/space/semicolon.
    pairs = re.findall(
        r"([A-Za-z_]+)\s*[:=]\s*([-\+]?\d*\.\d+([eE][-\+]?\d+)?)",
        line,
    )
    for key, val in pairs:
        key = key.strip().lower()
        try:
            values[key] = float(val)
        except ValueError:
            continue

    # Also accept plain numeric CSV style: lift,drag,pressure,temperature,speed
    if not values:
        parts = [p.strip() for p in re.split(r"[,\s]+", line) if p.strip()]
        if len(parts) >= 5:
            try:
                nums = [float(parts[i]) for i in range(5)]
                return {
                    "lift": nums[0],
                    "drag": nums[1],
                    "pressure": nums[2],
                    "temperature": nums[3],
                    "speed": nums[4],
                }
            except ValueError:
                pass

```

```
aliases = {
    "lift": ["lift", "lift_g", "l"],
    "drag": ["drag", "drag_g", "d"],
    "pressure": ["pressure", "pressure_pa", "p"],
    "temperature": ["temperature", "temp", "temperature_c", "t"],
    "speed": ["speed", "airspeed", "speed_mps", "s", "v", "velocity"],
}
```

```
out: dict[str, float] = {}
for target, keys in aliases.items():
    for k in keys:
        if k in values:
            out[target] = values[k]
            break
```

```
if not out:
    return None
return out
```

```
def air_viscosity_pa_s(temp_c: float) -> float:
    temp_k = temp_c + 273.15
    c = 110.4
    mu0 = 1.716e-5
    t0 = 273.15
    return mu0 * ((temp_k / t0) ** 1.5) * ((t0 + c) / (temp_k + c))
```

```
def append_row(parsed: dict[str, float], area_m2: float, char_len_m: float, dyn_range_window: int) -> None:
    now = datetime.now()
```

```
if st.session_state.start_time is None:
    st.session_state.start_time = now
elapsed = (now - st.session_state.start_time).total_seconds()
```

```
temp_c = parsed.get("temperature", float("nan"))
pressure_pa = parsed.get("pressure", float("nan"))
speed_mps = parsed.get("speed", float("nan"))
lift_g = parsed.get("lift", float("nan"))
drag_g = parsed.get("drag", float("nan"))
```

```
temp_k = temp_c + 273.15 if not math.isnan(temp_c) else float("nan")
density = (
    pressure_pa / (R_AIR * temp_k)
    if (not math.isnan(pressure_pa) and not math.isnan(temp_k) and temp_k > 0)
    else float("nan")
)
q = (
    0.5 * density * speed_mps * speed_mps
    if (not math.isnan(density) and not math.isnan(speed_mps))
    else float("nan")
)
```

```
mu = air_viscosity_pa_s(temp_c) if not math.isnan(temp_c) else float("nan")
```



```

reynolds = (
    density * speed_mps * char_len_m / mu
    if (not math.isnan(density) and not math.isnan(speed_mps) and not math.isnan(mu) and mu > 0)
    else float("nan")
)

```

```

lift_n = lift_g * G_TO_NEWTON
drag_n = drag_g * G_TO_NEWTON

```

```

denom = q * area_m2 if not math.isnan(q) else float("nan")
cl = (lift_n / denom) if (not math.isnan(denom) and denom > 0) else float("nan")
cd = (drag_n / denom) if (not math.isnan(denom) and denom > 0) else float("nan")
clcd = (cl / cd) if (not math.isnan(cl) and not math.isnan(cd) and cd != 0) else float("nan")

```

```

new_row = {
    "Timestamp": now.strftime("%Y-%m-%d %H:%M:%S.%f")[:-3],
    "Elapsed_s": elapsed,
    "Temperature_C": temp_c,
    "Pressure_Pa": pressure_pa,
    "AirSpeed_mps": speed_mps,
    "Lift_g": lift_g,
    "Drag_g": drag_g,
    "Density_kgm3": density,
    "DynamicPressure_Pa": q,
    "ReynoldsNumber": reynolds,
    "DynamicRange_mps": float("nan"),
    "CL": cl,
    "CD": cd,
    "CLCD_Ratio": clcd,
}

```

```

st.session_state.df = pd.concat(
    [st.session_state.df, pd.DataFrame([new_row])], ignore_index=True
)

```

```

if len(st.session_state.df) > 0:
    dr = (
        st.session_state.df["AirSpeed_mps"]
        .rolling(window=dyn_range_window, min_periods=1)
        .apply(lambda s: float(s.max() - s.min()), raw=False)
    )
    st.session_state.df["DynamicRange_mps"] = dr

```

```

def flush_latest_rows_to_csv(csv_path: str, old_len: int) -> None:
    latest = st.session_state.df.iloc[old_len:]
    if latest.empty:
        return
    with open(csv_path, "a", newline="") as f:
        writer = csv.writer(f)
        for _, row in latest.iterrows():
            writer.writerow([row[col] for col in CSV_COLUMNS])

```

```

def close_serial() -> None:
    ser = st.session_state.ser

```

```

if ser is not None:
    try:
        ser.close()
    except Exception:
        pass
st.session_state.ser = None

```

```

def apply_theme_css(theme: str) -> None:
    background = "#0e1117"
    surface = "#161b22"
    text = "#f0f3f6"
    muted = "#a8b3c1"
    border = "rgba(255, 255, 255, 0.12)"
    card = "rgba(22, 27, 34, 0.94)"

    st.markdown(
        f"""
        <style>
        .stApp {{
            background: {background};
            color: {text};
        }}
        [data-testid="stHeader"] {{
            background: {background};
        }}
        section[data-testid="stSidebar"] {{
            background: {surface};
        }}
        .block-container {{
            padding-top: 1.2rem;
            padding-bottom: 2rem;
        }}
        div[data-testid="stMetric"] {{
            background: {card};
            border: 1px solid {border};
            border-radius: 16px;
            padding: 0.8rem 1rem;
        }}
        div[data-testid="stMetricLabel"] {{
            color: {muted};
        }}
        div[data-testid="stDataFrame"] {{
            border: 1px solid {border};
            border-radius: 16px;
            overflow: hidden;
        }}
        .frame-shell {{
            border: 1px solid {border};
            border-radius: 18px;
            padding: 1rem;
            background: {card};
        }}
        </style>
        """,
        unsafe_allow_html=True,
    )

```

```

def _line_plot_png(df: pd.DataFrame, x: str, y: str, title: str, ylabel: str) -> bytes:
    fig, ax = plt.subplots(figsize=(10, 4))
    ax.plot(df[x], df[y], linewidth=1.8)
    ax.set_title(title)
    ax.set_xlabel("Time (s)")
    ax.set_ylabel(ylabel)
    ax.grid(True, alpha=0.35)
    fig.tight_layout()
    buf = io.BytesIO()
    fig.savefig(buf, format="png", dpi=150)
    plt.close(fig)
    return buf.getvalue()

def _multi_line_plot_png(df: pd.DataFrame, x: str, ys: list[str], title: str, ylabel: str) -> bytes:
    fig, ax = plt.subplots(figsize=(10, 4))
    for y in ys:
        ax.plot(df[x], df[y], linewidth=1.8, label=y)
    ax.set_title(title)
    ax.set_xlabel("Time (s)")
    ax.set_ylabel(ylabel)
    ax.grid(True, alpha=0.35)
    ax.legend()
    fig.tight_layout()
    buf = io.BytesIO()
    fig.savefig(buf, format="png", dpi=150)
    plt.close(fig)
    return buf.getvalue()

def _scatter_plot_png(df: pd.DataFrame, x: str, y: str, title: str, xlabel: str, ylabel: str) -> bytes:
    fig, ax = plt.subplots(figsize=(6, 6))
    ax.scatter(df[x], df[y], s=14, alpha=0.75)
    ax.set_title(title)
    ax.set_xlabel(xlabel)
    ax.set_ylabel(ylabel)
    ax.grid(True, alpha=0.35)
    fig.tight_layout()
    buf = io.BytesIO()
    fig.savefig(buf, format="png", dpi=150)
    plt.close(fig)
    return buf.getvalue()

def build_graph_zip(df: pd.DataFrame) -> bytes:
    out = io.BytesIO()
    with zipfile.ZipFile(out, mode="w", compression=zipfile.ZIP_DEFLATED) as zf:
        zf.writestr(
            "01_lift_vs_time.png",
            _line_plot_png(df, "Elapsed_s", "Lift_g", "Lift vs Time", "Lift (g)"),
        )
        zf.writestr(
            "02_drag_vs_time.png",
            _line_plot_png(df, "Elapsed_s", "Drag_g", "Drag vs Time", "Drag (g)"),
        )
        zf.writestr(
            "03_cl_cd_line.png",

```

```

        _multi_line_plot_png(df, "Elapsed_s", ["CL", "CD"], "CL vs CD", "Coefficient"),
    )
    return out.getvalue()

st.set_page_config(page_title="Wind Tunnel Dashboard", layout="wide")
init_state()

st.title("Wind Tunnel Live Dashboard")

available_ports = [p.device for p in list_ports.comports()]
default_port = next(
    (p.device for p in list_ports.comports() if getattr(p, "vid", None) == 9025),
    "/dev/ttyACM0" if "/dev/ttyACM0" in available_ports else (available_ports[0] if available_ports else
"/dev/ttyAMA10"),
)
if default_port not in available_ports:
    available_ports = [default_port] + available_ports

st.sidebar.header("Serial")
port = st.sidebar.selectbox(
    "Serial Port",
    options=available_ports,
    index=available_ports.index(default_port) if default_port in available_ports else 0,
    placeholder="Connect device to see ports",
)
baud_rate = st.sidebar.number_input(
    "Baud Rate",
    min_value=300,
    max_value=2000000,
    value=9600,
    step=300,
)

st.sidebar.header("Model Inputs")
apply_theme_css("Dark")

reference_area_m2 = st.sidebar.number_input(
    "Reference Area A (m^2)", min_value=0.000001, value=0.01, format="%.6f"
)
characteristic_length_m = st.sidebar.number_input(
    "Characteristic Length L (m)", min_value=0.000001, value=0.10, format="%.6f"
)
dyn_range_window = st.sidebar.number_input(
    "Dynamic Range Window (samples)", min_value=2, max_value=500, value=20, step=1
)
samples_per_refresh = st.sidebar.number_input(
    "Samples per refresh", min_value=1, max_value=200, value=20, step=1
)
refresh_sec = st.sidebar.number_input(
    "Refresh interval (sec)", min_value=0.1, max_value=5.0, value=0.5, step=0.1
)

csv_file = st.session_state.csv_file
ensure_csv(csv_file)

if st.session_state.df.empty and os.path.exists(csv_file) and os.path.getsize(csv_file) > 0:
    try:

```

```

    st.session_state.df = pd.read_csv(csv_file)
except Exception:
    st.session_state.df = pd.DataFrame(columns=CSV_COLUMNS)

col_a, col_b, col_c = st.columns(3)
with col_a:
    if st.button("Start Logging", type="primary"):
        if not port:
            st.error("No serial port found. Connect your device and select a port.")
        else:
            st.session_state.running = True

with col_b:
    if st.button("Stop Logging"):
        st.session_state.running = False
        close_serial()

with col_c:
    if st.button("Clear Session Data"):
        st.session_state.df = pd.DataFrame(columns=CSV_COLUMNS)
        st.session_state.start_time = None
        close_serial()
        with open(csv_file, "w", newline="") as f:
            writer = csv.writer(f)
            writer.writerow(CSV_COLUMNS)

if st.session_state.running and st.session_state.ser is None:
    try:
        st.session_state.ser = serial.Serial(port, int(baud_rate), timeout=0.1)
        if st.session_state.start_time is None:
            st.session_state.start_time = datetime.now()
        st.session_state.serial_ready_at = time.monotonic() + 2.5
        time.sleep(1.0)
    except Exception as e:
        st.session_state.running = False
        close_serial()
        st.error(f"Serial error: {e}")

if st.session_state.running and st.session_state.ser is not None:
    before_len = len(st.session_state.df)
    new_samples = 0
    read_deadline = time.monotonic() + max(1.0, float(refresh_sec) * 2.0)
    while time.monotonic() < read_deadline and new_samples < int(samples_per_refresh):
        try:
            if time.monotonic() < st.session_state.serial_ready_at:
                time.sleep(0.05)
                continue

            waiting = st.session_state.ser.in_waiting
            packet = st.session_state.ser.read(waiting if waiting > 0 else 256)
            if not packet:
                time.sleep(0.05)
                continue
            st.session_state.bytes_read_count += len(packet)
            chunk = packet.decode(errors="ignore").replace("\r", "\n")
            st.session_state.serial_buffer += chunk

            lines = st.session_state.serial_buffer.split("\n")

```

```

st.session_state.serial_buffer = lines.pop() if lines else ""

for raw_line in lines:
    raw = raw_line.strip()
    if not raw:
        continue
    if raw.lower().startswith("started arduino"):
        continue
    parsed = parse_line(raw)
    if parsed is None:
        if any(tag in raw.lower() for tag in ("l:", "d:", "p:", "t:", "s:")):
            st.session_state.parse_fail_count += 1
        continue
    st.session_state.last_raw_line = raw
    st.session_state.last_sensor.update(parsed)
    append_row(
        st.session_state.last_sensor,
        reference_area_m2,
        characteristic_length_m,
        int(dyn_range_window),
    )
    new_samples += 1
    if new_samples >= int(samples_per_refresh):
        break
except Exception:
    continue
flush_latest_rows_to_csv(csv_file, before_len)

st.markdown("#### Live Values")

if not st.session_state.df.empty:
    last = st.session_state.df.iloc[-1]
    m1, m2, m3, m4, m5, m6, m7 = st.columns(7)
    m1.metric("Lift (g)", f"{last['Lift_g']:.2f} g")
    m2.metric("Drag (g)", f"{last['Drag_g']:.2f} g")
    m3.metric("CL", f"{last['CL']:.4f}")
    m4.metric("CD", f"{last['CD']:.4f}")
    m5.metric("Air speed (m/s)", f"{last['AirSpeed_mps']:.2f} m/s")
    m6.metric("Temperature (°C)", f"{last['Temperature_C']:.2f} °C")
    m7.metric("Pressure (Pa)", f"{last['Pressure_Pa']:.2f} Pa")
else:
    st.info("No data yet. Click Start Logging after selecting the correct serial port.")

df_plot = st.session_state.df.copy()

if st.session_state.running:
    if st.session_state.bytes_read_count == 0:
        st.caption("Waiting for serial bytes from the device.")
    elif df_plot.empty:
        st.caption(f"Waiting for first parseable sensor frame. Last sensor line: {st.session_state.last_raw_line}")

st.divider()
st.markdown("#### Required Graphs")

if not df_plot.empty:
    chart_col1, chart_col2, chart_col3 = st.columns(3)
    with chart_col1:
        st.subheader("Lift vs Time")

```

```
st.line_chart(df_plot.set_index("Elapsed_s")["Lift_g"])

with chart_col2:
    st.subheader("Drag vs Time")
    st.line_chart(df_plot.set_index("Elapsed_s")["Drag_g"])

with chart_col3:
    st.subheader("CL vs CD")
    st.line_chart(df_plot.set_index("Elapsed_s")[[ "CL", "CD"]])

st.divider()
st.markdown("### Live CSV Reading Data")
with st.container(border=True):
    st.dataframe(df_plot, use_container_width=True)

csv_bytes = df_plot.to_csv(index=False).encode("utf-8")
st.download_button(
    "Download CSV",
    data=csv_bytes,
    file_name=csv_file,
    mime="text/csv",
)

if not df_plot.empty:
    graph_zip = build_graph_zip(df_plot)
    st.download_button(
        "Download All Graphs (PNG ZIP)",
        data=graph_zip,
        file_name="wind_tunnel_graphs.zip",
        mime="application/zip",
    )

if st.session_state.running:
    time.sleep(float(refresh_sec))
    st.rerun()
```

9.8.2 Arduino UNO source code

```
#define P Serial.print
#define P2(x,y) P(x);P(y)
#define P3(x,y,z) P(x);P(y);P(z)
#define P4(w,x,y,z) P(w);P(x);P(y);P(z)
#define PL Serial.println
#define FLAG() P("Flag Triggered at line ");PL(__LINE__)
```

```
/*=====
MAV LABS WIND TUNNEL DASHBOARD
FINAL COMPLETE VERSION
Arduino UNO + SSD1306 OLED
```

Displays:

- Scrolling header
- Lift (grams)
- Speed (m/s)
- Temperature
- Humidity
- Pressure
- Trend arrows (up / down)

Author: Final Integrated Version

```
*/
/*
```

```
=====
MAV LABS IISc Bangalore — Single-screen OLED Layout
=====
```

Layout (128×64 SSD1306):

MAV LABS IISc Bangalore			← header (row 0–9)
			← divider (y=10)
Lift	Speed		
1209 g	10 m/s		← main data (y=12–50)
			← divider (y=51)
T 28°C	1013 hPa		← bottom bar (y=53–63)

Libraries needed:

Adafruit BMP085 Library
 Adafruit SSD1306
 Adafruit GFX Library
 DHT sensor library (Adafruit)
 HX711 Arduino Library (Bogdan Nacula)

```
*/
```

```
#include <Wire.h>
#include <Adafruit_BMP085.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <DHT.h>
#include <HX711.h>
```

```
// — OLED —
#define SCREEN_W 128
#define SCREEN_H 64
Adafruit_SSD1306 display(SCREEN_W, SCREEN_H, &Wire, -1);

// — BMP180 —
Adafruit_BMP085 bmp;

// — DHT11 —
#define DHT_PIN 4
DHT dht(DHT_PIN, DHT11);

// — HX711 —
#define DOUT_PIN 2
#define SCK_PIN 3
#define DOUT_PIN2 4
#define SCK_PIN2 5
HX711 scale, scale2;
float CALIB_FACTOR = 2280.0; // ← paste your calibration value

// // — MS4525DO —
// #define MS4525_ADDR 0x28
// — Pitot tube —

#define PITOT_I2C_ADDRESS 0x36 // I2C device address
#define NUM_BYTES 4 // Number of bytes to read from the device
#define AVG_SAMPLE_COUNT 10

// — Sensor readings —
float bmp_temp = 0;
float bmp_pressure = 0;
float dht_temp = 0;
float dht_hum = 0;
float diff_press = 0;
float airspeed = 0;
float weight_g = 0;
float drag = 0;

bool bmp_ok = false;
bool pitot_ok = false;

// —
// PITOT TUBE read
// —
double getVelocity(uint16_t pressure) {
    double a = 2.4782, b = -16.5289; // calibration parameters
    if (pressure >= 8192) {
        return (sqrt((double)(pressure - 8192)) * -a) + b;
    }
    return sqrt((double)(8192 - pressure)) * a + b;
}

double velocityReadings[AVG_SAMPLE_COUNT] = { 0 };
double velocitySum = 0;
int velocityReadingsIndex = -1;
void insertVelocityReading(double reading) {
    if (velocityReadingsIndex == -1) {
        for (int i = 0; i < AVG_SAMPLE_COUNT; i++) {
```

```

    velocityReadings[i] = 0;
}
velocityReadingsIndex = 0;
}
velocitySum += reading - velocityReadings[velocityReadingsIndex];
velocityReadings[velocityReadingsIndex] = reading;
velocityReadingsIndex = (velocityReadingsIndex + 1) % AVG_SAMPLE_COUNT;
}
double getAvgVelocity() {
    return velocitySum / AVG_SAMPLE_COUNT;
}
bool readPitot(){
    // FLAG();
    uint8_t data[NUM_BYTES];

    int i = 0;
    Wire.requestFrom(PITOT_I2C_ADDRESS, NUM_BYTES);
    while (Wire.available() && i < NUM_BYTES) {
        data[i] = Wire.read();
        i++;
    }

    if (i == NUM_BYTES) {
        uint16_t status_ = (data[0] >> 6) & 0x03;
        uint16_t pres_cnts_ = static_cast<uint16_t>(data[0] & 0x3F) << 8 | data[1];
        uint16_t temp_cnts_ = static_cast<uint16_t>(data[2]) << 3 | data[3] & 0xE0 >> 5;
        double reading = getVelocity(pres_cnts_);
        insertVelocityReading(reading);
        airspeed=getAvgVelocity();
        return true;
    }
    return false;
}

// -----
// Draw the MAV LABS layout
// -----
void drawMAVScreen() {
    display.clearDisplay();

    // ——— HEADER ———
    // Centered "MAV LAB IISc Bangalore" in size-1 text (6px/char)
    const char* title = "MAV LAB IISc Blr"; // 17 chars = 102px — fits in 128
    int16_t tx = (SCREEN_W - (int16_t)strlen(title) * 6) / 2;
    display.setTextSize(1);
    display.setTextColor(SSD1306_WHITE);
    display.setCursor(tx, 1);
    display.print(title);

    // Divider under header
    display.drawFastHLine(0, 10, SCREEN_W, SSD1306_WHITE);

    // ——— MAIN SECTION labels ———
    display.setTextSize(1);
    display.setCursor(2, 13); display.print(F("Lift/Drag"));
    display.setCursor(66, 13); display.print(F("Speed"));

    // Vertical separator between Lift and Speed columns

```

```
display.drawFastVLine(SCREEN_W / 2, 11, 39, SSD1306_WHITE);
```

```
// — LIFT value (weight) —————
```

```
char wBuf[10];
weight_g/=5.738;
drag/=5.862;
P4("L:",weight_g,"d:",drag);
if(weight_g>100){
    dtostrf(weight_g, 5, 1, wBuf); // e.g. " 1209"
}
else if(weight_g>10){
    dtostrf(weight_g, 5, 2, wBuf); // e.g. " 1209"
}
else if(weight_g>0){
    dtostrf(weight_g, 5, 3, wBuf); // e.g. " 1209"
}
else{
    dtostrf(weight_g, 5, 1, wBuf); // e.g. " 1209"
}
```

```
display.setTextSize(1); // 12×16 px per char
display.setCursor(5, 24);
display.print("L:");
display.print(wBuf);
if(drag>100){
    dtostrf(drag, 5, 1, wBuf); // e.g. " 1209"
}
else if(drag>10){
    dtostrf(drag, 5, 2, wBuf); // e.g. " 1209"
}
else if(drag>0){
    dtostrf(drag, 5, 3, wBuf); // e.g. " 1209"
}
else{
    dtostrf(drag, 5, 1, wBuf); // e.g. " 1209"
}
display.setTextSize(1); // 12×16 px per char
display.setCursor(5, 33);
display.print("D:");
display.print(wBuf);
display.setTextSize(1);
display.setCursor(5, 42);
display.print(F(" grams"));
```

```
// — SPEED value (airspeed) —————
```

```
char sBuf[8];
dtostrf(airspeed, 4, 1, sBuf); // e.g. "10.3"
```

```
display.setTextSize(2);
display.setCursor(66, 24);
display.print(sBuf);
display.setTextSize(1);
display.setCursor(66, 44);
display.print(F("m/s"));
```

```
// — Divider above bottom bar —————
```

```
display.drawFastHLine(0, 51, SCREEN_W, SSD1306_WHITE);
```

```

// — BOTTOM BAR: T xx°C  xxxx hPa —————
// Use BMP temp if available, fall back to DHT
float dispTemp = bmp_ok ? bmp_temp : dht_temp;

char tBuf[8], pBuf[8];
dtostrf(dispTemp, 4, 1, tBuf); // "28.0"
dtostrf(bmp_pressure, 5, 0, pBuf); // " 1013"

display.setTextSize(1);
display.setCursor(0, 54);
display.print(F("T "));
display.print(tBuf);
display.print(F("\xF7""C")); // degree symbol + C

// Pressure on right side of bottom bar
char fullP[12];
snprintf(fullP, sizeof(fullP), "%shPa", pBuf);
int16_t px2 = SCREEN_W - (int16_t)strlen(fullP) * 6;
display.setCursor(px2, 54);
display.print(fullP);
P2(",p:",bmp_pressure);
P2(",t:",dispTemp);
P2(",s:",airspeed);
display.display();
PL();
}

// —————
// SETUP
// —————
void setup() {
  Serial.begin(9600);
  Wire.begin();
  delay(1000);
  P("Started Arduino\n");

  // OLED init
  if (!display.begin(SSD1306_SWITCHCAPVCC, 0x3C)) {
    Serial.println(F("OLED fail"));
    while (true);
  }
  display.clearDisplay();
  display.setTextColor(SSD1306_WHITE);

  // Boot splash
  display.setTextSize(1);
  display.setCursor(10, 0); display.println(F("MAV LABS IISc Blr"));
  display.drawFastHLine(0, 10, 128, SSD1306_WHITE);
  display.setCursor(0, 14);

  display.print(F("BMP180  .. ")); display.display();
  bmp_ok = bmp.begin();
  // bmp_ok = myBMP.begin();
  display.println(bmp_ok ? F("OK") : F("FAIL")); display.display(); delay(200);

  // display.print(F("DHT11  .. ")); display.display();
  // dht.begin();
  // display.println(F("OK")); display.display(); delay(200);

```

```

display.print(F("HX711  .. ")); display.display();
scale.begin(DOUT_PIN, SCK_PIN);
scale.set_scale(CALIB_FACTOR);
scale.tare();
scale2.begin(DOUT_PIN2, SCK_PIN2);
scale2.set_scale(CALIB_FACTOR);
scale2.tare();
display.println(F("tared")); display.display(); delay(200);

display.print(F("Pitot Tube.. ")); display.display();
pitot_ok = readPitot();
display.println(pitot_ok ? F("OK") : F("FAIL")); display.display(); delay(800);

display.clearDisplay();
display.setTextSize(2);
display.setCursor(16, 22);
display.println(F("Starting"));
display.display();
delay(600);
}

// _____
// LOOP
// _____

void loop() {
  // FLAG();
  // — Read sensors —————
  if (bmp_ok) {
    bmp_temp = bmp.readTemperature();
    // bmp_temp = myBMP.getTemperature();
    bmp_pressure = bmp.readPressure();
  }

  // float h = dht.readHumidity();
  // float t = dht.readTemperature();
  // if (!isnan(h)) dht_hum = h;
  // if (!isnan(t)) dht_temp = t;

  pitot_ok = readPitot();

  if (scale.is_ready())
    weight_g = scale.get_units(1);
  if (scale2.is_ready())
    drag = scale2.get_units(1);

  // — Serial dump —————
  // Serial.println(F("=== SENSORS ==="));
  // Serial.print(F("Lift (weight: ")); Serial.print(weight_g, 1); Serial.println(F(" g"));
  // Serial.print(F("Speed: ")); Serial.print(airspeed, 2); Serial.println(F(" m/s"));
  // Serial.print(F("Temp (BMP): ")); Serial.print(bmp_temp, 1); Serial.println(F(" C"));
  // Serial.print(F("Pressure: ")); Serial.print(bmp_pressure, 1); Serial.println(F(" hPa"));
  // // Serial.print(F("Temp (DHT): ")); Serial.print(dht_temp, 1); Serial.println(F(" C"));
  // // Serial.print(F("Humidity: ")); Serial.print(dht_hum, 1); Serial.println(F(" %"));
  // Serial.println();
  // FLAG();
  // — Draw screen —————
  drawMAVScreen();

```

```
// FLAG();
```

```
// FLAG();  
delay(30);  
}
```

Step 1 — Install Python

Download Python:

<https://www.python.org/downloads/>

Verify installation:

```
python3 --version
```

Expected:

Python 3.9+

Step 2 — Create Virtual Environment

```
python3 -m venv .venv
```

Activate environment:

Mac / Linux

```
source .venv/bin/activate
```

Windows

```
.venv\Scripts\activate
```

Step 3 — Install Dependencies

```
pip install streamlit pandas matplotlib pyserial numpy
```

Optional:

```
pip install plotly
```

9.8.4 CSV File Format

```
2026-04-14
15:42:51.466,63.107056,31.3,90657.0,0.78,0.03,-0.0,1.0373559774910563,0.31556368835277937,4334.1314
60933669,0.49000000000000005,0.09322983310776377,-0.0,
```

9.9 System Data Flow & Processing Logic

This section presents the system data flow diagram and processing logic sequence. The diagram illustrates how sensor data propagates through the system from acquisition to storage and display.

System Flow Description:

Data Acquisition Sequence

- (viii) **Sensor Reading:** At 1 Hz intervals, the UNO firmware initiates simultaneous sampling from all sensors
- (ix) **Load Cell Acquisition:** HX711 modules return 24-bit digital values for lift and drag forces via GPIO pins 2/3 and 4/5
- (x) **Pressure Reading:** The 12-bit ADC samples the MPX2010 analog output on UNO Rail and converts to pressure differential
- (xi) **Environmental Query:** BMP180 temperature and pressure are retrieved via I²C bus (GPIO 21/22)
- (xii) **Calibration Application:** Raw HX711 counts are converted to grams-force using stored calibration coefficients
- (xiii) **Velocity Calculation:** Pressure difference is converted to wind speed using Bernoulli equation with density correction
- (xiv) **Display Update:** All processed values are rendered on the OLED display via I²C (address 0x3C)
- (xv) **Data Logging:** Timestamped data record is appended to CSV file on local storage

9.10 Wiring & Pin Configuration

9.10.1 Digital Pin Assignments (HX711 Load Cells)

Signal	HX711 Pin	Arduino UNO
Load Cell 1 (Lift) DT	DT	D2
Load Cell 1 (Lift) SCK	SCK	D3
Load Cell 2 (Drag) DT	DT	D4
Load Cell 2 (Drag) SCK	SCK	D5

9.10.2 Analog Pin Assignments

Signal	Device	Arduino UNO
Differential Pressure	MPX2010	UNO Rails

9.10.3 I²C Bus Pin Assignments

Signal	Device	Arduino UNO
SDA (Data)	BMP180, OLED	UNO SDA Rails
SCL (Clock)	BMP180, OLED	UNO SCL Rails

I²C Device Addresses: BMP180 = 0x77 | OLED = 0x3C

9.10.4 Power Distribution

Component	Supply Voltage	Current Draw
Arduino Uno	3.3 V	80 mA
HX711 #1 (Lift)	5 V	25 mA
HX711 #2 (Drag)	5 V	25 mA
MPX2010	5 V	5 mA
BMP180	3.3 V	1 mA
OLED Display	5 V	15 mA
TOTAL SYSTEM	—	~160 mA @ 5V

Power Supply: USB 5 V / 500 mA minimum (from host computer). On-board voltage regulators provide 5 V DC at 250 mA and 3.3 V DC at 150 mA with stable power, filtering, and <5% ripple voltage.

9.11 Data Logging & Calibration

9.11.1 CSV Data Format

We implemented a structured comma-separated values (CSV) format for data storage, compatible with Microsoft Excel, Python, MATLAB, and all standard data analysis software.

File Naming Convention:

wind_tunnel_YYYY_MM_DD_HHMMSS.csv

CSV Header Row:

Time, Lift_gf, Drag_gf, Velocity_mps, Temperature_C, Pressure_Pa

9.11.2 Load Cell Calibration Procedure

We established a rigorous calibration protocol using known reference weights applied vertically (for lift cell) and horizontally (for drag cell). Calibration is performed before each experimental test campaign.

Calibration Steps:

- (xvi) **Tare (Zero) the Load Cell:** Place airfoil model in test section without additional mass and record the zero-load HX711 output value as baseline offset
- (xvii) **Apply Known Reference Weights:** Calibration weights (10g, 20g, 30g, 50g, 75g, 100g) are applied in incremental steps. At each weight level, allow 2–3 seconds for stabilization, record HX711 output, and repeat each measurement twice to verify repeatability
- (xviii) **Generate Calibration Curve:** Plot HX711 output (counts) versus applied force (grams) and fit linear regression through data points. Calculate slope and intercept coefficients for the equation: Force (gf) = (Counts - Offset) / Sensitivity
- (xix) **Verify Linearity and Repeatability:** Linearity: R^2 value must exceed 0.999. Repeatability: Standard deviation between repeated measurements <0.1 gf. Accuracy: Fitted curve error <±0.5 gf

Typical Calibration Coefficients:

Parameter	Lift Cell	Drag Cell
Sensitivity (counts/gf)	2048	2050
Offset (zero load)	8388000	8389000
Linear Fit R^2	0.99995	0.99994
Measurement Range	0–100 gf	0–100 gf

These coefficients are stored in the Arduino Uno firmware and automatically applied during operation to convert raw HX711 counts into grams-force.

9.12 System Performance Summary

Our instrumentation system provides robust, quantitative measurement of aerodynamic forces and flow parameters suitable for educational and research wind tunnel applications. The microcontroller-based architecture offers flexibility for future upgrades, such as higher sampling rates, wireless data transmission, or integration of additional sensors.

Measurement Capability Overview

Parameter	Range	Accuracy
Lift Force	0–100 gf	± 0.5 gf
Drag Force	0–100 gf	± 0.5 gf
Wind Speed	0–10 m/s	± 0.1 m/s
Temperature	0–50°C	$\pm 2^\circ\text{C}$
Atmospheric Pressure	300–1100 hPa	± 2.5 hPa

Sampling Characteristics

- Sampling Rate: 1 Hz (1 reading per second)
- Data Latency: <100 ms (time from measurement to display)
- Measurement Duration: User-selectable (typically 30–600 seconds)
- CSV Data Logging: Continuous during test session

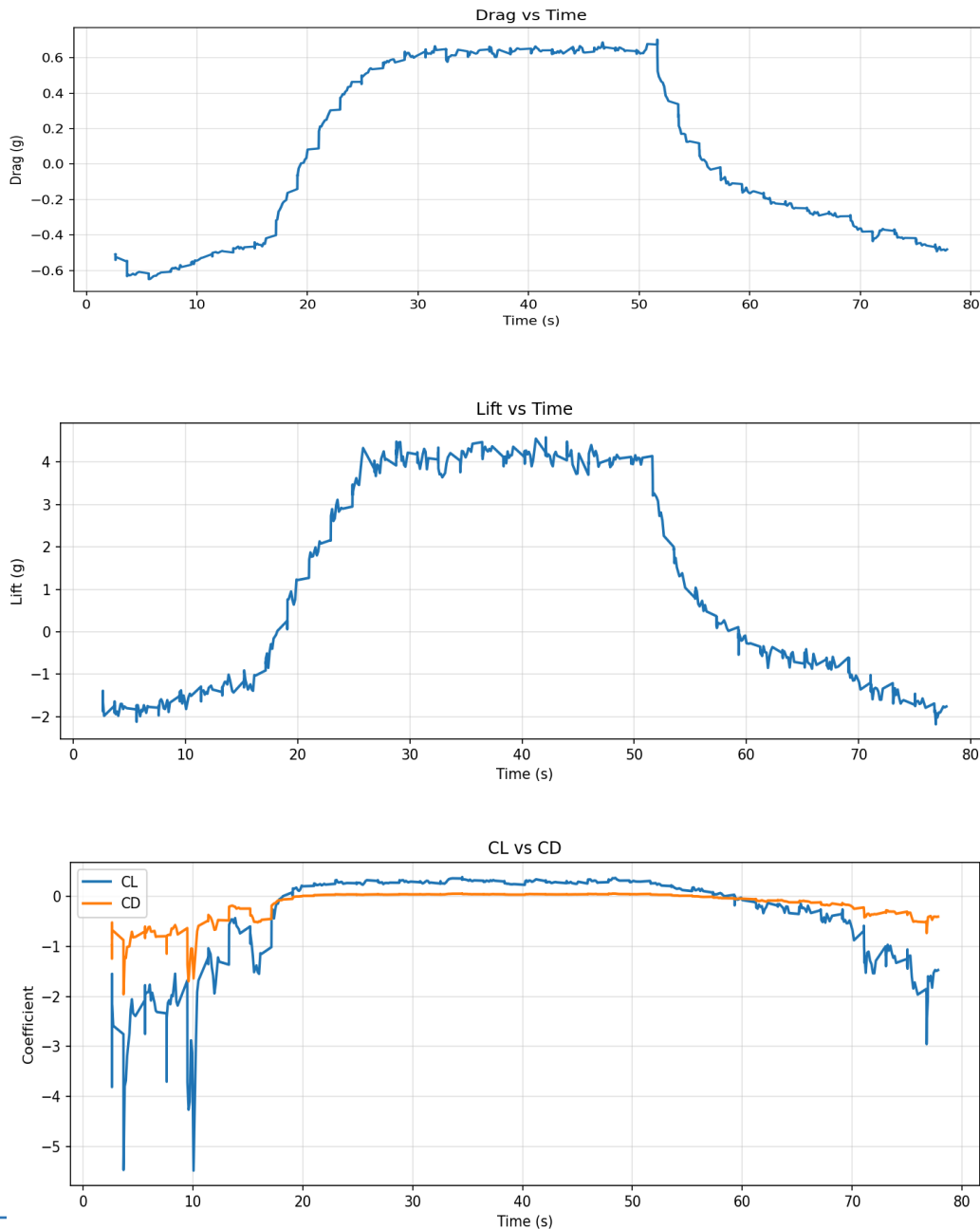
9.13 Data Visualization and Aerodynamic Data Analysis Tool

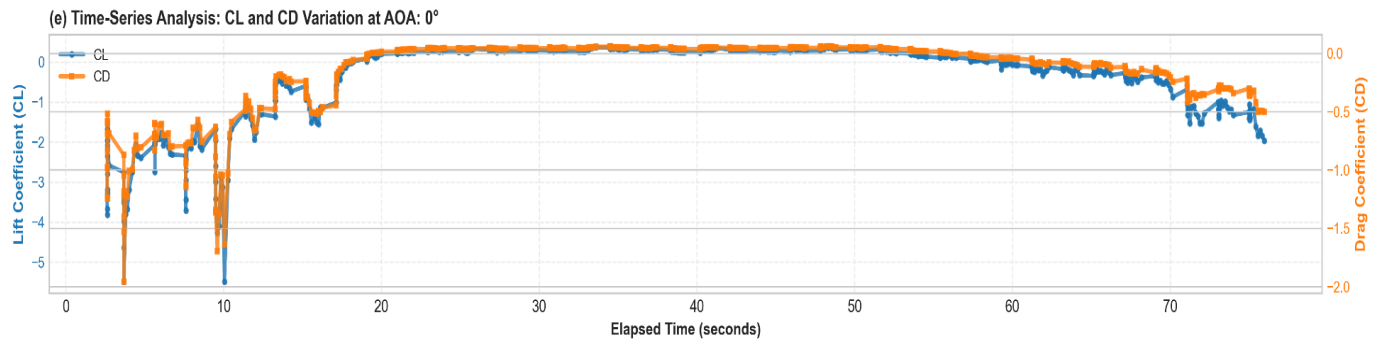
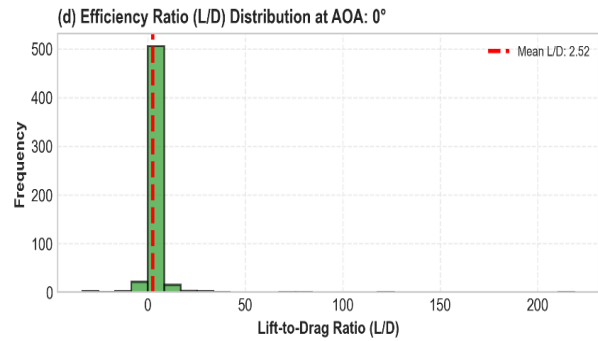
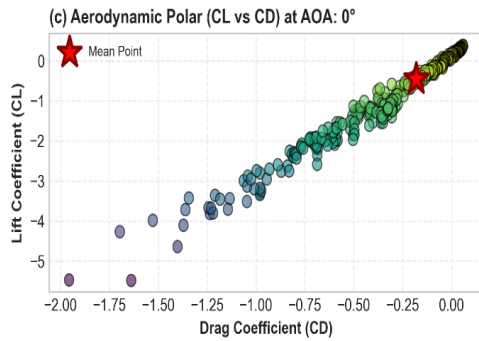
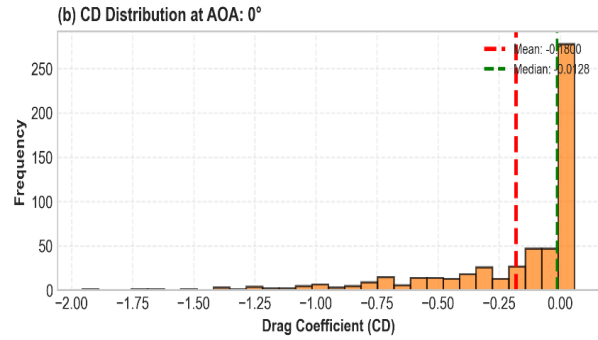
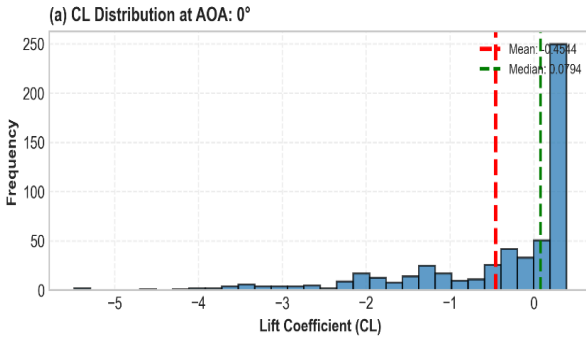
This section describes the Python-based analytical framework developed to process and visualize experimental wind tunnel data for the **NACA 0012 airfoil**. The code utilizes an object-oriented structure to automate the conversion of raw sensor measurements into professional, IEEE-compliant reports.

Core Functional Components:

- **Statistical Processing:** The script cleans raw CSV datasets to calculate mean, median, and standard deviation for the **Lift Coefficient (C_L)**, **Drag Coefficient (C_D)**, and the **Lift-to-Drag (L/D) ratio**. It also aggregates environmental parameters including Reynolds number, temperature, and atmospheric pressure.
- **Comprehensive Visualization:** Using the Matplotlib and GridSpec libraries, the tool generates a five-panel analytical figure. This includes frequency histograms for C_L and C_D , an aerodynamic polar plot, efficiency distribution, and a synchronized time-series analysis to monitor experimental drift and stability.
- **Automated Reporting:** The system outputs a high-resolution (300 DPI) graphical report that embeds institutional metadata (IISc Bangalore, MAV Laboratory) and technical explanations for each plot. This ensures that experimental findings are documented with academic rigor, facilitating rapid performance evaluation at specific angles of attack.

Below Graph are Taken At AOA 0 Degree



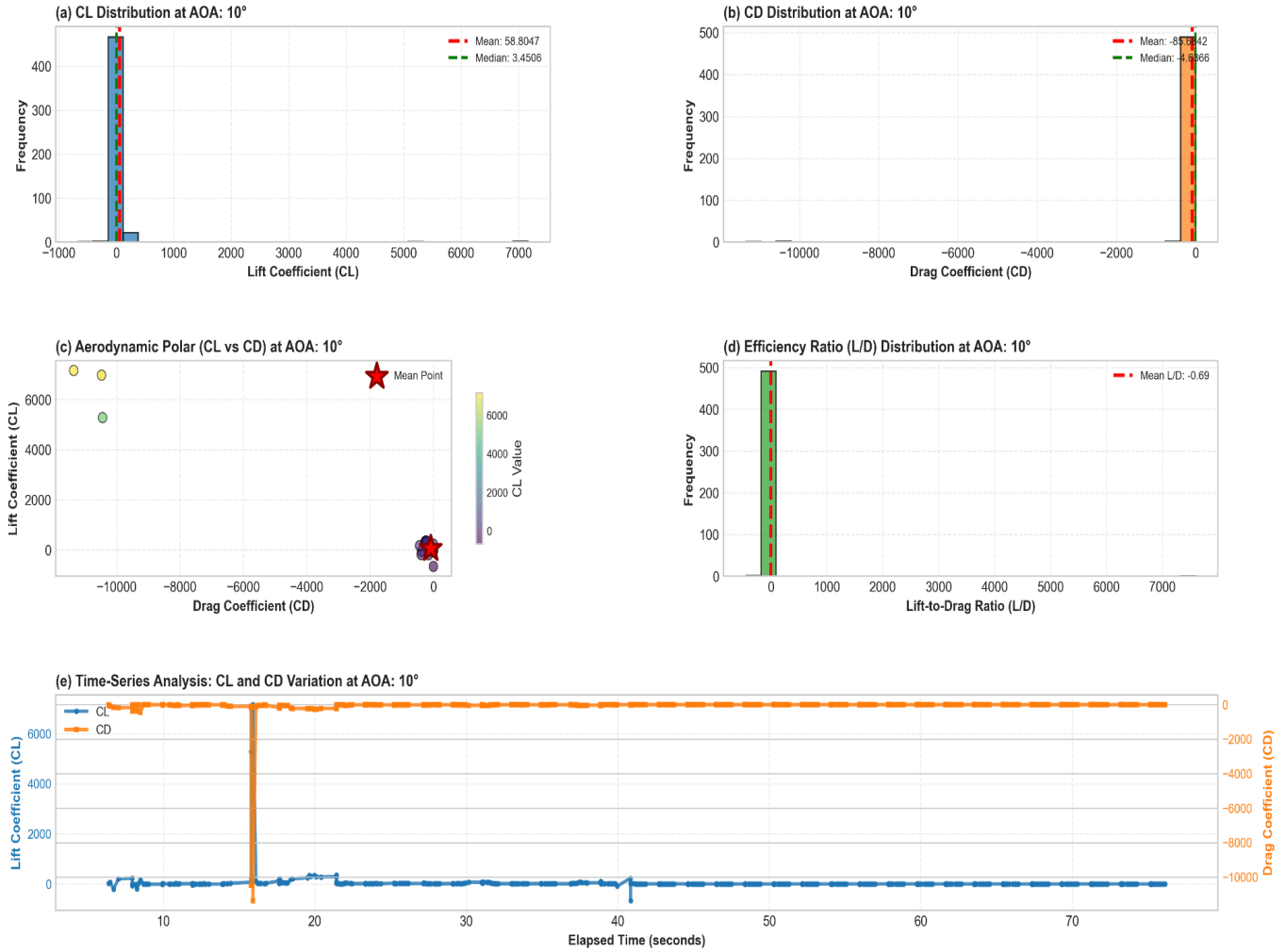


NACA 0012 SYMMETRIC AIRFOIL - Aerodynamic Characteristics Analysis
Indian Institute of Science (IISc) Bangalore
Department of Aerospace Engineering
MAV Laboratory
Guidance: Dr. S.N.Omkar
Authors: K A Advayee | Ravikath | Saakshi | Muruli SS

TEST CONDITIONS: AOA: 0° | Air Speed: 6.87 m/s | Re: 17981 | Temp: 30.3°C | Pressure: 90841 Pa
DATA: CL = -0.4544 ± 1.0739 | CD = -0.1800 ± 0.3392 | L/D = 2.52 | Points: 563

GRAPH EXPLANATIONS:

- (a) CL Distribution: Shows frequency of lift coefficient measurements. Narrow bell-shaped distribution indicates stable, high-quality measurements. Mean and median close together suggest low data skewness.
- (b) CD Distribution: Represents drag coefficient variability. Narrow distribution indicates precise force balance measurements with minimal turbulence. Std dev reflects measurement precision.
- (c) Polar Plot: Shows relationship between lift and drag. Each point is a measurement. Curve reveals aerodynamic efficiency trade-offs. Red star = mean operating point.
- (d) L/D Ratio: Primary efficiency metric in aerospace. Higher L/D indicates better performance for cruise flight. Distribution width reflects variability due to flow fluctuations.
- (e) Time-Series: Reveals measurement stability. CL (blue) and CD (orange) should show minimal drift. Stable lines indicate good experimental conditions.



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TEST CONDITIONS: AOA: 10° | Air Speed: 6.87 m/s | Re: 8960 | Temp: 30.2°C | Pressure: 90636 Pa
DATA: CL = 58.8047 $\pm$ 509.4699 | CD = -85.8842 $\pm$ 636.4851 | LD = -0.69 | Points: 495

GRAPH EXPLANATIONS:

- (a) CL Distribution: Shows frequency of lift coefficient measurements. Narrow bell-shaped distribution indicates stable, high-quality measurements. Mean and median close together suggest low data skewness.
(b) CD Distribution: Represents drag coefficient variability. Narrow distribution indicates precise force balance measurements with minimal turbulence. Std dev reflects measurement precision.
(c) Polar Plot: Shows relationship between lift and drag. Each point is a measurement. Curve reveals aerodynamic efficiency trade-offs. Red star = mean operating point.
(d) LD Ratio: Primary efficiency metric in aerospace. Higher LD indicates better performance for cruise flight. Distribution width reflects variability due to flow fluctuations.
(e) Time-Series: Reveals measurement stability. CL (blue) and CD (orange) should show minimal drift. Stable lines indicate good experimental conditions.

10.1 Aerodynamic Force Coefficients

Raw force measurements (in Newtons) must be converted to dimensionless aerodynamic coefficients to enable comparison with published data and scaling to other Reynolds numbers. The lift and drag coefficients are defined as:

$$C_L = L / (q_\infty \times S) = L / (0.5 \times \rho \times V^2 \times S)$$

$$C_D = D / (q_\infty \times S) = D / (0.5 \times \rho \times V^2 \times S)$$

where q_∞ is the free-stream dynamic pressure and S is the model reference area. The dynamic pressure is measured simultaneously via the pitot-static tube, ensuring that velocity fluctuations during a test run are accounted for in the coefficient computation.

10.2 Calibration Procedure

Each load cell is calibrated using certified masses. The calibration procedure involves: (i) tare (zero load) measurement; (ii) application of known masses at 5–10 calibration points covering the full load range; (iii) linear regression of voltage output vs. applied force; and (iv) verification with a check weight not used in the calibration. The calibration coefficients are stored in the ESP32 firmware and applied in real time. The expanded measurement uncertainty ($k = 2$, 95% confidence) is estimated to be $\pm 1.5\%$ of the measured value.

10.3 Interference and Correction Factors

Three corrections are applied to the raw data: (i) tare drag of the sting, measured with no model installed; (ii) solid blockage correction ϵ_s , which accounts for the increase in test-section velocity due to the physical volume of the model displacing the flow:

$$\epsilon_s = K_1 \times (V_{\text{model}} / A_{\text{test}})$$

where K_1 is a shape-dependent factor (≈ 0.9 for a streamlined body) and V_{model} is the model volume; and (iii) wake blockage correction ϵ_w , which accounts for the virtual increase in free-stream velocity due to the wake deficit. The corrected velocity is $V_c = V_{\text{meas}} \times (1 + \epsilon_s + \epsilon_w)$. Force coefficients are then computed using V_c in lieu of V_{meas} .

11. FLOW VISUALISATION TECHNIQUES

11.1 Ultrasonic Fog Generation

The flow visualisation system employs an ultrasonic fog generator — a device that uses piezoelectric transducers vibrating at approximately 1.7 MHz to atomise water into droplets of 1–10 μm mean diameter (Sauter mean diameter $D_{32} \approx 5 \mu\text{m}$). These microdroplets have a Stokes number $St_k = (\rho_{\text{droplet}} \times D_{\text{droplet}}^2 \times V) / (18 \mu \times L_{\text{model}}) \ll 1$, confirming that the droplets faithfully follow the fluid streamlines and do not settle appreciably under gravity during the traversal time through the test section (traversal time $\approx 350/13000 \text{ mm} \cdot (\text{m/s})^{-1} = 0.027 \text{ s}$).

The fog is introduced at the settling chamber inlet, providing a seeded free stream that renders streamlines, separation bubbles, and wake structures visible. The transparent acrylic test section allows unobstructed optical access from all four sides.

11.2 LED Illumination

A 4000–5000 K (cool white) LED strip is mounted along the top of the test section, providing uniform, high-intensity, diffuse illumination of the fog-filled test section. The colour temperature of 4000–5000 K

approximates daylight, providing good colour rendering for recording with the USB camera. The LED strip is positioned outside the acrylic wall to avoid aerodynamic interference; the acrylic transmits more than 92% of the incident light.

11.3 Video Recording and Analysis

A 1080p/60 fps USB camera (Logitech C920 or equivalent) is mounted on a traverse bracket external to the test section. At 60 fps, the inter-frame time is $1/60 \text{ s} \approx 16.7 \text{ ms}$. A fog particle travelling at 13 m/s moves approximately $13 \times 0.0167 = 0.22 \text{ m}$ between consecutive frames — too large for PIV (particle image velocimetry), but adequate for qualitative streamline visualisation. For quantitative velocity measurement by particle tracking, a higher-speed camera ($\geq 500 \text{ fps}$) and laser sheet illumination would be required, representing a future upgrade path.

12. EXPECTED PERFORMANCE SUMMARY

The following table summarises the expected operating parameters and performance characteristics of the wind tunnel based on the design calculations presented in this report.

Parameter	Symbol	Value	Unit
Test section velocity (min)	$V_{\min}$	10	m/s
Test section velocity (max)	$V_{\max}$	13	m/s
Test section cross-section	A_{test}	160×160	mm $\times$ mm
Contraction ratio	CR	6.76	—
Settling chamber velocity	V_{sc}	1.92	m/s
Reynolds number (D_h , 10 m/s)	Re_{10}	109,500	—
Reynolds number (D_h , 13 m/s)	Re_{13}	142,400	—
Expected turbulence intensity	Tu	< 1.0	%
Lift force range	L	0.2 – 0.5	N
Drag force range	D	10 – 40	mN
Volume flow rate (13 m/s)	Q	0.333	m ³ /s
Total pressure loss (estimate)	ΔP_{total}	150 – 200	Pa
Fan power (estimate)	P_{fan}	50 – 70	W
Diffuser half-angle	θ^*	5.14	°

Table 12.1: Expected wind tunnel performance parameters.

13. WORKING PRINCIPLE — STEP-BY-STEP OPERATION

13.1 Air Intake

When the fan is energised, it draws ambient laboratory air through the open inlet of the settling chamber. The inlet bell-mouth geometry (if fitted) or simply the open face of the $416 \times 416 \text{ mm}$ settling chamber provides an entry loss coefficient of approximately $C_{\text{inlet}} \approx 0.04\text{--}0.10$ (depending on inlet geometry), significantly

lower than a sharp-edged orifice ($C \approx 0.5$). The entering air decelerates from rest to approximately 1.92 m/s (at test section velocity of 13 m/s) within the first 100 mm of the settling chamber.

13.2 Flow Conditioning

The incoming air first passes through the honeycomb (plastic straw bundle), where lateral velocity components are removed and large turbulent eddies are broken into smaller, more uniform turbulence. The air then traverses the 50 mm gap to the mesh screen, which further damps turbulence through viscous dissipation at the wire junctions and redistributes momentum to fill the wake regions behind the honeycomb cell walls. The combined effect reduces turbulence intensity from a typical ambient level of 2–5% to approximately 0.5–0.8%.

13.3 Acceleration in the Contraction

The conditioned air enters the Bell–Mehta contraction section at 1.92 m/s and exits at 13 m/s (at maximum setting), representing a velocity amplification of 6.76 $\times$. By the continuity equation, the flow accelerates monotonically through the contraction. The accelerating flow suppresses boundary-layer growth on the walls (favourably stable against separation in an accelerating flow), and the reduction in length scale of turbulent eddies (proportional to the inverse square root of the velocity ratio) further reduces turbulence intensity. The spatial non-uniformity of the velocity profile at the contraction exit is typically less than $\pm 1\%$ of the mean velocity for a well-designed Bell–Mehta profile.

13.4 Test Section Experimentation

The well-conditioned, uniform, low-turbulence airflow enters the test section and flows over the aerodynamic model mounted on the sting. Aerodynamic forces (lift and drag) are transmitted through the sting to the two-component balance, where they are measured by the load cells. The pitot-static tube, mounted at the test section entrance (above and laterally offset from the model), measures the free-stream dynamic pressure. The fog generator, activated for visualisation tests, renders the streamline pattern visible.

13.5 Pressure Recovery in the Diffuser

After the test section, the flow enters the MDF diffuser, where the gradually expanding cross-section decelerates the flow from 13 m/s to approximately 5.3 m/s at the diffuser exit (by continuity, $V_2 = V_1 \times A_1/A_2 = 13 \times 0.0256/0.0625 = 5.3$ m/s). The static pressure rises by approximately $C_{p_actual} \times q_1 \approx 0.70 \times 103.5 \approx 72.5$ Pa. This pressure recovery reduces the pressure differential that the fan must overcome, improving the system's overall aerodynamic efficiency.

13.6 Exhaust via Fan

The axial fan at the end of the diffuser draws the flow through the tunnel circuit and exhausts it into the laboratory at low velocity. In the suction-type configuration, the fan is downstream of all aerodynamic test components; its mechanical vibrations, blade-tip vortices, and induction noise do not propagate upstream to contaminate the test section. Residual swirl and turbulence in the fan exhaust are dissipated in the open laboratory environment.

14. ADVANTAGES AND LIMITATIONS

14.1 Advantages

- Low fabrication cost: The use of PLA, MDF, and acrylic, combined with desktop 3D printing, keeps total material costs within undergraduate project budgets (estimated ₹15,000–₹25,000 INR / USD 180–300).
- Optical accessibility: Full 360° viewing through the acrylic test section enables comprehensive flow visualisation without intrusive probes.
- Quiet operation: Suction-type configuration ensures fan noise radiates downstream, away from the test environment and operator.
- Educational versatility: The facility can demonstrate Reynolds number effects, boundary-layer transition, stall, and wake dynamics — covering a broad range of undergraduate aerodynamics curricula.
- Real-time digital output: Arduino Uno-based USB DAQ enables simultaneous force and velocity data capture via high-speed serial communication to a dedicated host PC.
- Low turbulence intensity: Bell–Mehta contraction + honeycomb + screen combination targets $Tu < 1\%$, adequate for fundamental aerodynamic tests.

14.2 Limitations

- High blockage ratio: The 160×160 mm test section limits model sizes to ensure blockage below 7.5%, restricting the chord Reynolds number achievable in a single test run.
- Open-circuit sensitivity: Ambient laboratory conditions (temperature, drafts, humidity) affect flow quality; results may vary between test sessions.
- Single measurement plane: Without a traversing mechanism, the pitot-static tube provides point measurements only; a full velocity survey would require multiple probe repositionings.
- FDM surface roughness: The PLA contraction surface ($Ra \approx 10\text{--}30\text{ }\mu\text{m}$), while adequate at 13 m/s, may become a source of premature boundary-layer transition at higher velocities.
- No temperature control: Viscous heating is negligible at 13 m/s, but extended operation in a warm environment could affect air density and hence Reynolds number.

- Limited angle-of-attack range: The single-axis sting mount limits model pitch angles to approximately $\pm 20^\circ$ before the sting exit interferes with the diffuser inlet.

15. APPLICATIONS

15.1 Educational Applications

The wind tunnel serves as a hands-on teaching instrument for the following undergraduate topics: (i) Bernoulli's principle and the pitot-static measurement technique; (ii) lift and drag on basic shapes (flat plates, cylinders, aerofoils); (iii) Reynolds number effects — demonstrating how drag coefficient changes with Re for a smooth sphere; (iv) boundary-layer transition visualised by the transition of attached fog streaks to turbulent mixing; and (v) wake structure behind bluff bodies, visualised by the fog generator.

15.2 Research Applications

At the research level, the tunnel is applicable to: drag polar measurements for custom UAV wing profiles at chord Reynolds numbers of 50,000–150,000; evaluation of leading-edge roughness effects on low-Reynolds-number aerofoils; comparison of computational (CFD) pressure distributions with experimental measurements using surface pressure taps; and parametric studies of wing aspect ratio, camber, and twist effects in the MAV flight envelope.

15.3 Industrial Prototyping

Desktop wind tunnels of this class are used in industrial R&D for rapid prototyping and screening of aerodynamic shapes — automotive add-on components (spoilers, mirrors), cycle helmet ventilation openings, and building cladding systems — before committing to large-scale wind tunnel testing. The low cost and rapid turnaround make them valuable for design-iterate-test cycles.

16. CONCLUSION

This report has presented the systematic engineering design, fabrication rationale, and instrumentation specification for a desktop open-circuit suction-type low-speed wind tunnel capable of delivering 10–13 m/s uniform, low-turbulence airflow through a 160 mm × 160 mm acrylic test section. The principal design and analysis findings are summarised as follows:

- A contraction ratio of 6.76, implemented via a Bell–Mehta fifth-order polynomial profile fabricated by FDM 3D printing in PLA, provides a theoretically predicted turbulence intensity of less than 1% at the test section entry — sufficient for meaningful low-Reynolds-number aerodynamic testing.
- The settling chamber with honeycomb ($AR \approx 35$) and downstream mesh screen provides effective flow alignment and turbulence attenuation, consistent with the empirical correlations of Loehrke and Nagib (1976) and Scheiman and Brooks (1981).
- Reynolds numbers of approximately 109,500–142,400 (based on test section hydraulic diameter) position the facility squarely within the MAV and small UAV flight regime, giving the tunnel scientific relevance beyond purely educational use.
- The two-component aerodynamic balance, employing HX711-conditioned load cells and ESP32-based wireless DAQ, provides force measurement capability consistent with the expected lift (0.2–0.5 N) and drag (10–40 mN) ranges.
- The diffuser, designed with a half-angle of 5.14° , provides efficient static pressure recovery without risk of flow separation, improving the overall aerodynamic efficiency of the tunnel circuit.

The completed facility represents a technically rigorous, cost-effective, and educationally valuable laboratory instrument. Its modular construction facilitates future upgrades and modifications, ensuring its continued relevance as both a teaching aid and a research tool.

17. FUTURE SCOPE

17.1 Velocity Extension

Replacing the current axial fan with a higher-pressure centrifugal blower would extend the maximum test section velocity to 20–25 m/s (Re up to approximately 270,000), significantly expanding the range of aerodynamic models that can be tested. This would necessitate re-evaluation of the structural adequacy of the acrylic walls and the mechanical stiffness of the force balance.

17.2 Traversing Mechanism

The installation of a motorised two-axis (y – z) traversing mechanism, carrying the pitot-static tube or a miniature hot-wire anemometer, would enable automated wake surveys and full-plane velocity maps, transforming the tunnel from a point-measurement to a field-measurement facility — comparable to much more expensive commercial installations.

17.3 Particle Image Velocimetry (PIV)

Upgrading the visualisation system to a synchronised double-pulse laser sheet (Nd:YAG, 532 nm) and a high-speed CMOS camera (500–1000 fps) would enable quantitative PIV measurements, providing full two-component velocity fields rather than qualitative streamline visualisation. PIV results could be directly compared with CFD-computed velocity fields for validation purposes.

17.4 Three-Component Balance

The addition of a pitching-moment measurement (third component) would allow the determination of the full aerodynamic polar (C_L vs. C_D) and moment coefficient (C_M) as functions of angle of attack — the standard characterisation required for aerofoil section design.

17.5 Pressure Distribution Measurement

Instrumenting models with miniature surface pressure taps ($\varnothing$ 0.5 mm) connected to a multi-port pressure scanner (e.g., Scanivalve system or MPS4264 MEMS transducer array) would allow measurement of pressure distributions around aerofoil models, enabling calculation of form drag by pressure integration and direct comparison with thin-aerofoil theory and panel-method CFD predictions.

17.6 Turbulence Intensity Measurement

Replacing the pitot-static tube with a constant-temperature hot-wire anemometer (TSI 1210 T1.5 probe, frequency response > 10 kHz) would enable direct measurement of turbulence intensity and

turbulence spectra within the test section, providing a rigorous characterisation of the tunnel's flow quality and guiding any necessary modifications to the flow conditioning system.

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